



ŘEDITELSTVÍ SILNIC A DÁLNIC ČR

NDIC - DATEX II Situation Publication - Weather

Release 1.1rev0

Národní Dopravní Informační Centrum

2018-04-29

CONTENTS

Identifier and version:

uri
cz-ndic_d2-weather-v1.1

Format
1.1

Specification
1.1rev0

Updated
2018-04-29

Format type:

Publication
DATEX II v2.3/SituationPublication

Extends
—

Publisher and author:

Publisher
NDIC

Author
Jan Vlčinský, Petr Bureš

The document describes a data format used for publishing weather related information impacting the drivers.

INTRODUCTION

This document shall be used by publisher and / or subscriber. Subscriber shall use it for format suitability evaluation and for possible implementation of received NDIC data import into own information system. Publisher shall use it as a specification of data to be published format and for possible implementation of data export from own information system into NDIC.

Here you can find:

- conceptual description of published information
- information about location referencing methods used
- W3C XML schema of the format
- data sample(s)

Following topics are out of scope of this document and if present, shall be considered as informative only:

- transfer protocol incl. access point, life cycle etc.
- related data formats
- typical data origin
- typical data use

1.1 General terms

Following terms are used in this document:

format

description of data structure for transferring some kind of information

protocol

description of data transfer process and rules

NDIC

national traffic information centre

DATEX II model

class model for describing information related to the road traffic. The model in form of UML is platform independent.

One of specific platforms for expressing the model is XML.

The model is extensive (it consists of few thousands of structural items representing term incl. definition) For reference data (UML class model in form of Enterprise Architect EAP file, XML schema etc.) see [?]. For additional information see [?].

(DATEX II) profile

simplified class model, still sufficient for specific needs. Profile is created from model by selecting only specific parts.

(DATEX II) extension

extended DATEX II model, adding data structures missing in standard model but needed for specific needs.

B level extension

Extension of DATEX II model, which is backward compatible with the original DATEX II model, even though it contains additional structures. [?]

C level extension

Extension of DATEX II model, which is not backward compatible with the original DATEX II model.

publication

an element of DATEX II model, serving as an envelope for transmitted content of certain kind. Every sent DATEX II message contains exactly one element of publication type. There are various types of publications, for example *SituationPublication* [?], *ElaboratedDataPublication* [?], *PredefinedLocationsPublication* [?] etc.

(W3C) XML schema

language, which allows definition of rules for a XML document structure.

XSD

XML document containing XML schema. Sometimes it is synonym for XML schema itself.

uri

uniform resource identifier, text, used as unique identifier. In this document, every format has it's own uri.

CONCEPTS

Described format allows publishing weather related information impacting driving conditions for a set of predefined locations. This information belongs to category of traffic information, it is not published in predefined intervals, the contents carry event type information, like “dense fog”.

2.1 Publication type SituationPublication

Format is based on DATEX II publication of type *SituationPublication* [?]. Profile uses only small subset of possible structures.

```
1 <?xml version="1.0" encoding="utf-8"?>
2 <d2LogicalModel modelBaseVersion="2"
3 xmlns="http://datex2.eu/schema/2/2_0"
4 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5 xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">
6
7   <exchange>
8     <supplierIdentification>
9       <country>cz</country>
10      <nationalIdentifier>NDIC</nationalIdentifier>
11    </supplierIdentification>
12  </exchange>
13  <payloadPublication lang="cs" xsi:type="SituationPublication">
14    <feedType>x-format:cz-ndic_d2-weather-v1.1</feedType>
15    <publicationTime>2017-07-15T04:27:59+02:00</publicationTime>
16    <publicationCreator>
17      <country>cz</country>
18      <nationalIdentifier>NDIC</nationalIdentifier>
19    </publicationCreator>
```

2.2 Event description (only TrafficElement/Conditions)

SituationPublication requires description by means of class *Situation*, which aggregates instances of class *SituationRecord*, namely descendants of class *Conditions*. Besides the *SituationRecord* it contains elements with information about service availability.

Element *situation* describes one particular event, more particular events means more *situation* elements. Since the situation described may evolve over time, the element is versioned and has unique identifier. Each *situation* element contains one or more *situationRecord* elements.

The *situationRecord* element may, in the element *situation*, occur several times (each element describing different aspect of the situation, i.e. accident in one record leading to a queue described in another record). The *situationRecord* element may evolve over time, so the element is versioned and has unique identifier.

```

20 <situation id="MeteoService15.1" version="5">
21   <situationVersionTime>2017-07-15T04:27:21+02:00</situationVersionTime>
22   <headerInformation>
23     <confidentiality>noRestriction</confidentiality>
24     <informationStatus>real</informationStatus>
25     <urgency>normalUrgency</urgency>
26   </headerInformation>
27   <situationRecord id="MeteoService15.1_PoorEnvironmentConditions"
28     version="5" xsi:type="PoorEnvironmentConditions">
29     <situationRecordCreationTime>2017-07-15T04:25:59+02:00</
↪ situationRecordCreationTime>
30     <situationRecordVersionTime>2017-07-15T04:27:21+02:00</
↪ situationRecordVersionTime>
31     <probabilityOfOccurrence>certain</probabilityOfOccurrence>
32     <source>
33       <sourceIdentification>ITSMTNDIC</sourceIdentification>
34       <sourceName>
35         <values>
36           <value lang="cs">Správci meteohlásek DI</value>
37         </values>
38       </sourceName>
39     </source>
40     <validity>
41       <validityStatus>definedByValidityTimeSpec</validityStatus>
42       <validityTimeSpecification>
43         <overallStartTime>2017-07-15T04:26:50+02:00</overallStartTime>
44         <overallEndTime>2017-07-15T04:46:50+02:00</overallEndTime>
45       </validityTimeSpecification>
46     </validity>
47     <generalPublicComment>
48       <comment>
49         <values>
50           <value lang="cs">Hustá mlha, viditelnost menší než 30 m,
51             Vysocanská</value>
52         </values>
53       </comment>
54     </generalPublicComment>
55     <groupOfLocations xsi:type="LocationByReference">
56       <predefinedLocationReference id="73D74C70-BFA8-408F-86CD-49B3689D9F5D"
57         targetClass="PredefinedLocation" version="1" />
58     </groupOfLocations>
59     <poorEnvironmentType>denseFog</poorEnvironmentType>
60     <visibility>
61       <minimumVisibilityDistance>
62         <integerMetreDistance>30</integerMetreDistance>
63       </minimumVisibilityDistance>
64     </visibility>
65   </situationRecord>

```

2.3 Location description (by predefined locations)

The position can be described by a point or a short linear section on the road network, given by the presence of a weather station. Since the position of the weather sensing sensors does not change, the location in this profile is described by a reference (id) to a predefined location set defined in different document. The reference points to a class instance *PredefinedLocation* from package *PredefinedLocationsPublication*. Therefore, the location is not described directly in the file.

Location of weather information is determined by reference to a set of predefined locations. To look up given location, it is necessary to obtain publication *PredefinedLocationsPublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *PredefinedLocation*), with the same identifier and version.

The location for which the weather information is valid is specified by the *groupOfLocations* element re-typed to *xsi:type="LocationByReference"*, that includes element *predefinedLocationReference* with the attribute *targetClass="PredefinedLocation"*.

```
<groupOfLocations xsi:type="LocationByReference">
  <predefinedLocationReference id="73D74C70-BFA8-408F-86CD-49B3689D9F5D"
    targetClass="PredefinedLocation" version="1" />
</groupOfLocations>
```

The position description methods are further described in the chapter “Location description”.

TIME VALIDITY

Time validity is stated independently for every single *SituationRecord* described in a situation and is defined by a time interval between two time instants. The class *Validity* is used with an instance of class *OverallPeriod* in the role of *validityTimeSpecification* and elements *overallStartTime* and *overallStopTime* to define start and end of the interval.

Apart from methods mentioned above, no other methods for time validity specification are allowed (e.g. hours of validity or days of validity).

```
40     <validity>
41       <validityStatus>definedByValidityTimeSpec</validityStatus>
42       <validityTimeSpecification>
43         <overallStartTime>2017-07-15T04:26:50+02:00</overallStartTime>
44         <overallEndTime>2017-07-15T04:46:50+02:00</overallEndTime>
45       </validityTimeSpecification>
46     </validity>
```

LOCATION REFERENCING

The format describes routes exclusively by a reference to set of predefined locations (class *PredefinedLocation*) in publication *PredefinedLocationsPublication* [?].

Location referencing methods are out of scope of this document and are described elsewhere.

APPENDIX A: MESSAGE EXAMPLES

5.1 visibility.xml

Publication contains exactly one message about dense fog and visibility of 30 meters.

```
<?xml version="1.0" encoding="utf-8"?>
<d2LogicalModel modelBaseVersion="2"
xmlns="http://datex2.eu/schema/2/2_0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://datex2.eu/schema/2/2_0 ../schema/schema.xsd">

  <exchange>
    <supplierIdentification>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </supplierIdentification>
  </exchange>
  <payloadPublication lang="cs" xsi:type="SituationPublication">
    <feedType>x-format:cz-ndic_d2-weather-v1.1</feedType>
    <publicationTime>2017-07-15T04:27:59+02:00</publicationTime>
    <publicationCreator>
      <country>cz</country>
      <nationalIdentifier>NDIC</nationalIdentifier>
    </publicationCreator>
    <situation id="MeteoService15.1" version="5">
      <situationVersionTime>2017-07-15T04:27:21+02:00</situationVersionTime>
      <headerInformation>
        <confidentiality>noRestriction</confidentiality>
        <informationStatus>real</informationStatus>
        <urgency>normalUrgency</urgency>
      </headerInformation>
      <situationRecord id="MeteoService15.1_PoorEnvironmentConditions"
version="5" xsi:type="PoorEnvironmentConditions">
        <situationRecordCreationTime>2017-07-15T04:25:59+02:00</
↪situationRecordCreationTime>
        <situationRecordVersionTime>2017-07-15T04:27:21+02:00</
↪situationRecordVersionTime>
        <probabilityOfOccurrence>certain</probabilityOfOccurrence>
        <source>
          <sourceIdentification>ITSMTNDIC</sourceIdentification>
          <sourceName>
            <values>
              <value lang="cs">Správci meteohlásek DI</value>
            </values>
          </sourceName>
        </source>
      </situationRecord>
    </situation>
  </payloadPublication>

```

(continues on next page)

(continued from previous page)

```

</source>
<validity>
  <validityStatus>definedByValidityTimeSpec</validityStatus>
  <validityTimeSpecification>
    <overallStartTime>2017-07-15T04:26:50+02:00</overallStartTime>
    <overallEndTime>2017-07-15T04:46:50+02:00</overallEndTime>
  </validityTimeSpecification>
</validity>
<generalPublicComment>
  <comment>
    <values>
      <value lang="cs">Hustá mlha, viditelnost menší než 30 m,
        Vysocanská</value>
    </values>
  </comment>
</generalPublicComment>
<groupOfLocations xsi:type="LocationByReference">
  <predefinedLocationReference id="73D74C70-BFA8-408F-86CD-49B3689D9F5D"
    targetClass="PredefinedLocation" version="1" />
</groupOfLocations>
<poorEnvironmentType>denseFog</poorEnvironmentType>
<visibility>
  <minimumVisibilityDistance>
    <integerMetreDistance>30</integerMetreDistance>
  </minimumVisibilityDistance>
</visibility>
</situationRecord>
</situation>
</payloadPublication>
</d2LogicalModel>

```

5.1.1 Root element and publication envelope

Root element *d2LogicalModel* declares, that this message is using DATEX II.

Element *exchange* contains provider identifier.

Element *payloadPublication* is an envelope for actual information payload containing general information and one or more elements *situation*, (in our case there is one).

Attribute *xsi:type="SituationPublication"* in element *payloadPublication* determines, that this envelope is of type Situation publication.

Element *feedType* states publication schema by its uri.

Element *publicationTime* states publication time of this publication (file) as a whole.

Element *publicationCreator* identifies creator of the publication.

5.1.2 Description of one specific situation

Element *situation* describes a particular event, more particular events means more *situation* elements. Each *situation* element may contain one or more individual sub-messages, elements *situationRecord*. In this case occurs in the element *situation* just once.

Element *situationVersionTime* sets the time of the publication of this version of the situation.

Element *headerInformation* describes how to handle the information described.

Event description (what happened)

The attribute *xsi:type="PoorEnvironmentConditions"* of element *situationRecord* defines the event type. In DATEX II model [?] it can be verified, that given attribute is descendant of a class *Conditions*.

From this class, elements *poorEnvironmentType* are used with value of *denseFog*. Element *minimumVisibilityDistance* specifies visibility being 30 meters.

Time validity

Estimated time validity is given in a structure *validity* by values of validity interval start *overallStartTime* and validity interval end *overallEndTime*.

Event location

Event location is determined by reference to a set of predefined locations. To look up given location, it is necessary to obtain publication *PredefinedLocationsPublication*, related to this source of information and inside find location of identical type (by *targetClass*, here *PredefinedLocation*), with the same identifier and version.

The location for which the weather related event is valid is specified by the *groupOfLocations* element. It always has *xsi:type="LocationByReference"* and includes element *predefinedLocationReference* with the attribute *targetClass="PredefinedLocation"*.

The attribute *id="73D74C70-BFA8-408F-86CD-49B3689D9F5D"* refers to a predefined location with a given identifier, and the attribute *version='0'* specifies the version of the location description.

Additional content

The document states time of its first version in *situationRecordCreationTime* and time of current version in *situationRecordVersionTime* for each *situationRecord*.

Element *source* lists the the creator of this information, who is responsible for its accuracy.

Element *generalPublicComment* contains a general description of the situation, mostly only in Czech language. This is a free text description as provided by the situation creator.

APPENDIX B: W3C XML SCHEMA

XML structure of a transmitted message is defined by following W3C XML schema:

6.1 *schema.xsd*

```

1 <?xml version="1.0" encoding="utf-8" standalone="no"?>
2 <xs:schema elementFormDefault="qualified" attributeFormDefault="unqualified"
  ↳ xmlns:D2LogicalModel="http://datex2.eu/schema/2/2_0" version="2.3" targetNamespace=
  ↳ "http://datex2.eu/schema/2/2_0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
3   <xs:complexType name="_ExtensionType">
4     <xs:sequence>
5       <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs=
  ↳ "unbounded" />
6     </xs:sequence>
7   </xs:complexType>
8   <xs:complexType name="_PredefinedLocationVersionedReference">
9     <xs:complexContent>
10      <xs:extension base="D2LogicalModel:VersionedReference">
11        <xs:attribute name="targetClass" use="required" fixed="PredefinedLocation" />
12      </xs:extension>
13    </xs:complexContent>
14  </xs:complexType>
15  <xs:complexType name="_PredefinedNonOrderedLocationGroupVersionedReference">
16    <xs:complexContent>
17      <xs:extension base="D2LogicalModel:VersionedReference">
18        <xs:attribute name="targetClass" use="required" fixed=
  ↳ "PredefinedNonOrderedLocationGroup" />
19      </xs:extension>
20    </xs:complexContent>
21  </xs:complexType>
22  <xs:simpleType name="AngleInDegrees">
23    <xs:annotation>
24      <xs:documentation>An integer number representing an angle in whole degrees.
  ↳ between 0 and 359.</xs:documentation>
25    </xs:annotation>
26    <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
27  </xs:simpleType>
28  <xs:complexType name="ApplicationRateValue">
29    <xs:annotation>
30      <xs:documentation>A measured or calculated value of the application rate of a
  ↳ substance.</xs:documentation>
31    </xs:annotation>
32    <xs:complexContent>
33      <xs:extension base="D2LogicalModel:DataValue">

```

(continues on next page)

(continued from previous page)

```

34     <xs:sequence>
35       <xs:element name="applicationRate" type=
↪ "D2LogicalModel:IntensityKilogramsPerSquareMetre" minOccurs="1" maxOccurs="1">
36         <xs:annotation>
37           <xs:documentation>A value of the rate of application of a substance_
↪ expressed in kilogrammes per square metre.</xs:documentation>
38         </xs:annotation>
39       </xs:element>
40       <xs:element name="applicationRateValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
41     </xs:sequence>
42   </xs:extension>
43 </xs:complexContent>
44 </xs:complexType>
45 <xs:complexType name="AxleFlowValue">
46   <xs:annotation>
47     <xs:documentation>A measured or calculated value of the flow rate of vehicle_
↪ axles.</xs:documentation>
48   </xs:annotation>
49   <xs:complexContent>
50     <xs:extension base="D2LogicalModel:DataValue">
51       <xs:sequence>
52         <xs:element name="axleFlowRate" type="D2LogicalModel:AxlesPerHour" _
↪ minOccurs="1" maxOccurs="1">
53           <xs:annotation>
54             <xs:documentation>A value of the flow rate of vehicle axles expressed_
↪ in axles per hour.</xs:documentation>
55           </xs:annotation>
56         </xs:element>
57         <xs:element name="axleFlowValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
58       </xs:sequence>
59     </xs:extension>
60   </xs:complexContent>
61 </xs:complexType>
62 <xs:simpleType name="AxlesPerHour">
63   <xs:annotation>
64     <xs:documentation>Vehicle axles per hour.</xs:documentation>
65   </xs:annotation>
66   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
67 </xs:simpleType>
68 <xs:simpleType name="Boolean">
69   <xs:annotation>
70     <xs:documentation>Boolean has the value space required to support the_
↪ mathematical concept of binary-valued logic: {true, false}. </xs:documentation>
71   </xs:annotation>
72   <xs:restriction base="xs:boolean" />
73 </xs:simpleType>
74 <xs:complexType name="Cause" abstract="true">
75   <xs:annotation>
76     <xs:documentation>Contains details of the cause of a record within a situation</
↪ xs:documentation>
77   </xs:annotation>
78   <xs:sequence>
79     <xs:element name="causeExtension" type="D2LogicalModel:_ExtensionType" _
↪ minOccurs="0" />

```

(continues on next page)

(continued from previous page)

```

80     </xs:sequence>
81 </xs:complexType>
82 <xs:simpleType name="CauseTypeEnum">
83     <xs:annotation>
84         <xs:documentation>Types of causes of situations which are not managed or are
↳ off network.</xs:documentation>
85     </xs:annotation>
86     <xs:restriction base="xs:string">
87         <xs:enumeration value="accident">
88             <xs:annotation>
89                 <xs:documentation>Accident.</xs:documentation>
90             </xs:annotation>
91         </xs:enumeration>
92         <xs:enumeration value="congestion">
93             <xs:annotation>
94                 <xs:documentation>Traffic congestion.</xs:documentation>
95             </xs:annotation>
96         </xs:enumeration>
97         <xs:enumeration value="earlierAccident">
98             <xs:annotation>
99                 <xs:documentation>An earlier accident.</xs:documentation>
100            </xs:annotation>
101        </xs:enumeration>
102        <xs:enumeration value="earlierEvent">
103            <xs:annotation>
104                <xs:documentation>An earlier event.</xs:documentation>
105            </xs:annotation>
106        </xs:enumeration>
107        <xs:enumeration value="earlierIncident">
108            <xs:annotation>
109                <xs:documentation>An earlier incident.</xs:documentation>
110            </xs:annotation>
111        </xs:enumeration>
112        <xs:enumeration value="equipmentFailure">
113            <xs:annotation>
114                <xs:documentation>Failure of roadside equipment.</xs:documentation>
115            </xs:annotation>
116        </xs:enumeration>
117        <xs:enumeration value="excessiveHeat">
118            <xs:annotation>
119                <xs:documentation>Excessive heat.</xs:documentation>
120            </xs:annotation>
121        </xs:enumeration>
122        <xs:enumeration value="frost">
123            <xs:annotation>
124                <xs:documentation>Frost.</xs:documentation>
125            </xs:annotation>
126        </xs:enumeration>
127        <xs:enumeration value="holidayTraffic">
128            <xs:annotation>
129                <xs:documentation>Holiday traffic.</xs:documentation>
130            </xs:annotation>
131        </xs:enumeration>
132        <xs:enumeration value="infrastructureFailure">
133            <xs:annotation>
134                <xs:documentation>Failure of road infrastructure.</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

135     </xs:annotation>
136 </xs:enumeration>
137 <xs:enumeration value="largeNumbersOfVisitors">
138   <xs:annotation>
139     <xs:documentation>Large numbers of visitors.</xs:documentation>
140   </xs:annotation>
141 </xs:enumeration>
142 <xs:enumeration value="obstruction">
143   <xs:annotation>
144     <xs:documentation>Obstruction (of unspecified type) on the roadway.</
↪xs:documentation>
145   </xs:annotation>
146 </xs:enumeration>
147 <xs:enumeration value="pollutionAlert">
148   <xs:annotation>
149     <xs:documentation>An alert relating to dangerous levels of pollution.</
↪xs:documentation>
150   </xs:annotation>
151 </xs:enumeration>
152 <xs:enumeration value="poorWeather">
153   <xs:annotation>
154     <xs:documentation>Poor weather conditions.</xs:documentation>
155   </xs:annotation>
156 </xs:enumeration>
157 <xs:enumeration value="problemsAtBorderPost">
158   <xs:annotation>
159     <xs:documentation>Problems at the border crossing.</xs:documentation>
160   </xs:annotation>
161 </xs:enumeration>
162 <xs:enumeration value="problemsAtCustomPost">
163   <xs:annotation>
164     <xs:documentation>Problems at the customs post on the border.</
↪xs:documentation>
165   </xs:annotation>
166 </xs:enumeration>
167 <xs:enumeration value="problemsOnLocalRoads">
168   <xs:annotation>
169     <xs:documentation>Problems (of an unspecified nature) on the local roads.</
↪xs:documentation>
170   </xs:annotation>
171 </xs:enumeration>
172 <xs:enumeration value="radioactiveLeakAlert">
173   <xs:annotation>
174     <xs:documentation>Radioactive leak alert.</xs:documentation>
175   </xs:annotation>
176 </xs:enumeration>
177 <xs:enumeration value="roadsideEvent">
178   <xs:annotation>
179     <xs:documentation>A roadside event (of unspecified nature) whether planned,
↪or not.</xs:documentation>
180   </xs:annotation>
181 </xs:enumeration>
182 <xs:enumeration value="rubberNecking">
183   <xs:annotation>
184     <xs:documentation>Drivers being distracted and turning to look at an,
↪accident or other roadside event.</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

185     </xs:annotation>
186 </xs:enumeration>
187 <xs:enumeration value="securityIncident">
188   <xs:annotation>
189     <xs:documentation>A security incident.</xs:documentation>
190   </xs:annotation>
191 </xs:enumeration>
192 <xs:enumeration value="shearWeightOfTraffic">
193   <xs:annotation>
194     <xs:documentation>Shear weight of traffic.</xs:documentation>
195   </xs:annotation>
196 </xs:enumeration>
197 <xs:enumeration value="technicalProblems">
198   <xs:annotation>
199     <xs:documentation>Technical problems.</xs:documentation>
200   </xs:annotation>
201 </xs:enumeration>
202 <xs:enumeration value="terrorism">
203   <xs:annotation>
204     <xs:documentation>A terrorist incident.</xs:documentation>
205   </xs:annotation>
206 </xs:enumeration>
207 <xs:enumeration value="toxicCloudAlert">
208   <xs:annotation>
209     <xs:documentation>An alert relating to the release of toxic gases and/or
↪ particulates.</xs:documentation>
210   </xs:annotation>
211 </xs:enumeration>
212 <xs:enumeration value="vandalism">
213   <xs:annotation>
214     <xs:documentation>A vandalism incident.</xs:documentation>
215   </xs:annotation>
216 </xs:enumeration>
217 <xs:enumeration value="other">
218   <xs:annotation>
219     <xs:documentation>Other than as defined in this enumeration.</
↪ xs:documentation>
220   </xs:annotation>
221 </xs:enumeration>
222 </xs:restriction>
223 </xs:simpleType>
224 <xs:complexType name="Comment">
225   <xs:annotation>
226     <xs:documentation>A free text comment with an optional date/time stamp that can
↪ be used by the operator to convey un-coded observations/information.</
↪ xs:documentation>
227   </xs:annotation>
228   <xs:sequence>
229     <xs:element name="comment" type="D2LogicalModel:MultilingualString" minOccurs="1
↪ " maxOccurs="1">
230       <xs:annotation>
231         <xs:documentation>A free text comment that can be used by the operator to
↪ convey un-coded observations/information.</xs:documentation>
232       </xs:annotation>
233     </xs:element>
234     <xs:element name="commentDateTime" type="D2LogicalModel:DateTime" minOccurs="0"

```

(continues on next page)

(continued from previous page)

```

235     <xs:annotation>
236       <xs:documentation>The date/time at which the comment was made.</
237     </xs:annotation>
238   </xs:element>
239   <xs:element name="commentType" type="D2LogicalModel:CommentTypeEnum" minOccurs=
240     <xs:annotation>
241       <xs:documentation>A classification of the the type of comment.</
242     </xs:annotation>
243   </xs:element>
244   <xs:element name="commentExtension" type="D2LogicalModel:_ExtensionType"
245     </xs:sequence>
246 </xs:complexType>
247 <xs:simpleType name="CommentTypeEnum">
248   <xs:annotation>
249     <xs:documentation>Classification of comment types.</xs:documentation>
250   </xs:annotation>
251   <xs:restriction base="xs:string">
252     <xs:enumeration value="abnormalLoadMovementNote">
253       <xs:annotation>
254         <xs:documentation>A free text human oriented note describing details of
255       <xs:documentation>abnormal load movements associated with the SituationRecord.</xs:documentation>
256     </xs:annotation>
257     </xs:enumeration>
258     <xs:enumeration value="dataProcessingNote">
259       <xs:annotation>
260         <xs:documentation>A free text human oriented note describing the way the
261       <xs:documentation>information in the SituationRecord has been or should be processed.</
262     </xs:annotation>
263     </xs:enumeration>
264     <xs:enumeration value="description">
265       <xs:annotation>
266         <xs:documentation>A free text human oriented description of the situation
267       <xs:documentation>element defined by the SituationRecord.</xs:documentation>
268     </xs:annotation>
269     </xs:enumeration>
270     <xs:enumeration value="internalNote">
271       <xs:annotation>
272         <xs:documentation>A free text human oriented note that supports internal
273       <xs:documentation>traffic control operations relating to the situation element defined by the
274     </xs:documentation>SituationRecord.</xs:documentation>
275     </xs:annotation>
276     </xs:enumeration>
277     <xs:enumeration value="locationDescriptor">
278       <xs:annotation>
279         <xs:documentation>A free text human oriented description of the location of
280       <xs:documentation>the situation element defined by the SituationRecord.</xs:documentation>
281     </xs:annotation>
282     </xs:enumeration>
283     <xs:enumeration value="warning">
284       <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

279     <xs:documentation>A free text human oriented warning relating to the
↪ SituationRecord, such as advising the recipient that an advanced warning on VMS
↪ should be activated.</xs:documentation>
280     </xs:annotation>
281     </xs:enumeration>
282     <xs:enumeration value="other">
283     <xs:annotation>
284     <xs:documentation>Other than as defined in this enumeration.</
↪ xs:documentation>
285     </xs:annotation>
286     </xs:enumeration>
287 </xs:restriction>
288 </xs:simpleType>
289 <xs:simpleType name="ComputationMethodEnum">
290     <xs:annotation>
291     <xs:documentation>Types of computational methods used in deriving data values
↪ for data sets.</xs:documentation>
292     </xs:annotation>
293     <xs:restriction base="xs:string">
294     <xs:enumeration value="arithmeticAverageOfSamplesBasedOnAFixedNumberOfSamples">
295     <xs:annotation>
296     <xs:documentation>Arithmetic average of sample values based on a fixed
↪ number of samples.</xs:documentation>
297     </xs:annotation>
298     </xs:enumeration>
299     <xs:enumeration value="arithmeticAverageOfSamplesInATimePeriod">
300     <xs:annotation>
301     <xs:documentation>Arithmetic average of sample values in a time period.</
↪ xs:documentation>
302     </xs:annotation>
303     </xs:enumeration>
304     <xs:enumeration value="harmonicAverageOfSamplesInATimePeriod">
305     <xs:annotation>
306     <xs:documentation>Harmonic average of sample values in a time period.</
↪ xs:documentation>
307     </xs:annotation>
308     </xs:enumeration>
309     <xs:enumeration value="medianOfSamplesInATimePeriod">
310     <xs:annotation>
311     <xs:documentation>Median of sample values taken over a time period.</
↪ xs:documentation>
312     </xs:annotation>
313     </xs:enumeration>
314     <xs:enumeration value="movingAverageOfSamples">
315     <xs:annotation>
316     <xs:documentation>Moving average of sample values.</xs:documentation>
317     </xs:annotation>
318     </xs:enumeration>
319 </xs:restriction>
320 </xs:simpleType>
321 <xs:simpleType name="ConcentrationKilogramsPerCubicMetre">
322     <xs:annotation>
323     <xs:documentation>Concentration defined in kilograms per cubic metre
↪ (equivalent to grams per litre under standard conditions).</xs:documentation>
324     </xs:annotation>
325     <xs:restriction base="D2LogicalModel:Float" />

```

(continues on next page)

(continued from previous page)

```

326 </xs:simpleType>
327 <xs:complexType name="ConcentrationOfVehiclesValue">
328   <xs:annotation>
329     <xs:documentation>A measured or calculated value of the concentration of
↪ vehicles on a unit stretch of road in a given direction.</xs:documentation>
330   </xs:annotation>
331   <xs:complexContent>
332     <xs:extension base="D2LogicalModel:DataValue">
333       <xs:sequence>
334         <xs:element name="concentrationOfVehicles" type=
↪ "D2LogicalModel:ConcentrationVehiclesPerKilometre" minOccurs="1" maxOccurs="1">
335           <xs:annotation>
336             <xs:documentation>A value of traffic density expressed in the number of
↪ vehicles per kilometre of road.</xs:documentation>
337           </xs:annotation>
338         </xs:element>
339         <xs:element name="concentrationOfVehiclesValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
340       </xs:sequence>
341     </xs:extension>
342   </xs:complexContent>
343 </xs:complexType>
344 <xs:simpleType name="ConcentrationVehiclesPerKilometre">
345   <xs:annotation>
346     <xs:documentation>A measure of traffic density defined in number of vehicles
↪ per kilometre of road.</xs:documentation>
347   </xs:annotation>
348   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
349 </xs:simpleType>
350 <xs:complexType name="Conditions">
351   <xs:annotation>
352     <xs:documentation>Any conditions which have the potential to degrade normal
↪ driving conditions.</xs:documentation>
353   </xs:annotation>
354   <xs:complexContent>
355     <xs:extension base="D2LogicalModel:TrafficElement">
356       <xs:sequence>
357         <xs:element name="drivingConditionType" type=
↪ "D2LogicalModel:DrivingConditionTypeEnum" minOccurs="0" maxOccurs="1">
358           <xs:annotation>
359             <xs:documentation>Description of the driving conditions at the
↪ specified location.</xs:documentation>
360           </xs:annotation>
361         </xs:element>
362         <xs:element name="conditionsExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
363       </xs:sequence>
364     </xs:extension>
365   </xs:complexContent>
366 </xs:complexType>
367 <xs:simpleType name="ConfidentialityValueEnum">
368   <xs:annotation>
369     <xs:documentation>Values of confidentiality.</xs:documentation>
370   </xs:annotation>
371   <xs:restriction base="xs:string">
372     <xs:enumeration value="internalUse">

```

(continues on next page)

(continued from previous page)

```

373     <xs:annotation>
374       <xs:documentation>For internal use only of the recipient organisation.</
↪ <xs:documentation>
375     </xs:annotation>
376   </xs:enumeration>
377   <xs:enumeration value="noRestriction">
378     <xs:annotation>
379       <xs:documentation>No restriction on usage.</xs:documentation>
380     </xs:annotation>
381   </xs:enumeration>
382   <xs:enumeration value="restrictedToAuthorities">
383     <xs:annotation>
384       <xs:documentation>Restricted for use only by authorities.</xs:documentation>
385     </xs:annotation>
386   </xs:enumeration>
387   <xs:enumeration value="restrictedToAuthoritiesAndTrafficOperators">
388     <xs:annotation>
389       <xs:documentation>Restricted for use only by authorities and traffic_
↪ operators.</xs:documentation>
390     </xs:annotation>
391   </xs:enumeration>
392   <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndPublishers">
393     <xs:annotation>
394       <xs:documentation>Restricted for use only by authorities, traffic operators_
↪ and publishers (service providers).</xs:documentation>
395     </xs:annotation>
396   </xs:enumeration>
397   <xs:enumeration value="restrictedToAuthoritiesTrafficOperatorsAndVms">
398     <xs:annotation>
399       <xs:documentation>Restricted for use only by authorities, traffic operators,
↪ publishers (service providers) and variable message signs.</xs:documentation>
400     </xs:annotation>
401   </xs:enumeration>
402 </xs:restriction>
403 </xs:simpleType>
404 <xs:simpleType name="CountryEnum">
405   <xs:annotation>
406     <xs:documentation>List of countries.</xs:documentation>
407   </xs:annotation>
408   <xs:restriction base="xs:string">
409     <xs:enumeration value="at">
410       <xs:annotation>
411         <xs:documentation>Austria</xs:documentation>
412       </xs:annotation>
413     </xs:enumeration>
414     <xs:enumeration value="be">
415       <xs:annotation>
416         <xs:documentation>Belgium</xs:documentation>
417       </xs:annotation>
418     </xs:enumeration>
419     <xs:enumeration value="bg">
420       <xs:annotation>
421         <xs:documentation>Bulgaria</xs:documentation>
422       </xs:annotation>
423     </xs:enumeration>
424     <xs:enumeration value="ch">

```

(continues on next page)

(continued from previous page)

```

425     <xs:annotation>
426       <xs:documentation>Switzerland</xs:documentation>
427     </xs:annotation>
428   </xs:enumeration>
429   <xs:enumeration value="cs">
430     <xs:annotation>
431       <xs:documentation>Serbia and Montenegro</xs:documentation>
432     </xs:annotation>
433   </xs:enumeration>
434   <xs:enumeration value="cy">
435     <xs:annotation>
436       <xs:documentation>Cyprus</xs:documentation>
437     </xs:annotation>
438   </xs:enumeration>
439   <xs:enumeration value="cz">
440     <xs:annotation>
441       <xs:documentation>Czech Republic</xs:documentation>
442     </xs:annotation>
443   </xs:enumeration>
444   <xs:enumeration value="de">
445     <xs:annotation>
446       <xs:documentation>Germany</xs:documentation>
447     </xs:annotation>
448   </xs:enumeration>
449   <xs:enumeration value="dk">
450     <xs:annotation>
451       <xs:documentation>Denmark</xs:documentation>
452     </xs:annotation>
453   </xs:enumeration>
454   <xs:enumeration value="ee">
455     <xs:annotation>
456       <xs:documentation>Estonia</xs:documentation>
457     </xs:annotation>
458   </xs:enumeration>
459   <xs:enumeration value="es">
460     <xs:annotation>
461       <xs:documentation>Spain</xs:documentation>
462     </xs:annotation>
463   </xs:enumeration>
464   <xs:enumeration value="fi">
465     <xs:annotation>
466       <xs:documentation>Finland</xs:documentation>
467     </xs:annotation>
468   </xs:enumeration>
469   <xs:enumeration value="fo">
470     <xs:annotation>
471       <xs:documentation>Faroe Islands</xs:documentation>
472     </xs:annotation>
473   </xs:enumeration>
474   <xs:enumeration value="fr">
475     <xs:annotation>
476       <xs:documentation>France</xs:documentation>
477     </xs:annotation>
478   </xs:enumeration>
479   <xs:enumeration value="gb">
480     <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

481     <xs:documentation>Great Britain</xs:documentation>
482   </xs:annotation>
483 </xs:enumeration>
484 <xs:enumeration value="gg">
485   <xs:annotation>
486     <xs:documentation>Guernsey</xs:documentation>
487   </xs:annotation>
488 </xs:enumeration>
489 <xs:enumeration value="gi">
490   <xs:annotation>
491     <xs:documentation>Gibraltar</xs:documentation>
492   </xs:annotation>
493 </xs:enumeration>
494 <xs:enumeration value="gr">
495   <xs:annotation>
496     <xs:documentation>Greece</xs:documentation>
497   </xs:annotation>
498 </xs:enumeration>
499 <xs:enumeration value="hr">
500   <xs:annotation>
501     <xs:documentation>Croatia</xs:documentation>
502   </xs:annotation>
503 </xs:enumeration>
504 <xs:enumeration value="hu">
505   <xs:annotation>
506     <xs:documentation>Hungary</xs:documentation>
507   </xs:annotation>
508 </xs:enumeration>
509 <xs:enumeration value="ie">
510   <xs:annotation>
511     <xs:documentation>Ireland</xs:documentation>
512   </xs:annotation>
513 </xs:enumeration>
514 <xs:enumeration value="im">
515   <xs:annotation>
516     <xs:documentation>Isle Of Man</xs:documentation>
517   </xs:annotation>
518 </xs:enumeration>
519 <xs:enumeration value="is">
520   <xs:annotation>
521     <xs:documentation>Iceland</xs:documentation>
522   </xs:annotation>
523 </xs:enumeration>
524 <xs:enumeration value="it">
525   <xs:annotation>
526     <xs:documentation>Italy</xs:documentation>
527   </xs:annotation>
528 </xs:enumeration>
529 <xs:enumeration value="je">
530   <xs:annotation>
531     <xs:documentation>Jersey</xs:documentation>
532   </xs:annotation>
533 </xs:enumeration>
534 <xs:enumeration value="li">
535   <xs:annotation>
536     <xs:documentation>Lichtenstein</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

537     </xs:annotation>
538 </xs:enumeration>
539 <xs:enumeration value="lt">
540     <xs:annotation>
541         <xs:documentation>Lithuania</xs:documentation>
542     </xs:annotation>
543 </xs:enumeration>
544 <xs:enumeration value="lu">
545     <xs:annotation>
546         <xs:documentation>Luxembourg</xs:documentation>
547     </xs:annotation>
548 </xs:enumeration>
549 <xs:enumeration value="lv">
550     <xs:annotation>
551         <xs:documentation>Latvia</xs:documentation>
552     </xs:annotation>
553 </xs:enumeration>
554 <xs:enumeration value="ma">
555     <xs:annotation>
556         <xs:documentation>Morocco</xs:documentation>
557     </xs:annotation>
558 </xs:enumeration>
559 <xs:enumeration value="mc">
560     <xs:annotation>
561         <xs:documentation>Monaco</xs:documentation>
562     </xs:annotation>
563 </xs:enumeration>
564 <xs:enumeration value="mk">
565     <xs:annotation>
566         <xs:documentation>Macedonia</xs:documentation>
567     </xs:annotation>
568 </xs:enumeration>
569 <xs:enumeration value="mt">
570     <xs:annotation>
571         <xs:documentation>Malta</xs:documentation>
572     </xs:annotation>
573 </xs:enumeration>
574 <xs:enumeration value="nl">
575     <xs:annotation>
576         <xs:documentation>Netherlands</xs:documentation>
577     </xs:annotation>
578 </xs:enumeration>
579 <xs:enumeration value="no">
580     <xs:annotation>
581         <xs:documentation>Norway</xs:documentation>
582     </xs:annotation>
583 </xs:enumeration>
584 <xs:enumeration value="pl">
585     <xs:annotation>
586         <xs:documentation>Poland</xs:documentation>
587     </xs:annotation>
588 </xs:enumeration>
589 <xs:enumeration value="pt">
590     <xs:annotation>
591         <xs:documentation>Portugal</xs:documentation>
592     </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

593     </xs:enumeration>
594     <xs:enumeration value="ro">
595         <xs:annotation>
596             <xs:documentation>Romania</xs:documentation>
597         </xs:annotation>
598     </xs:enumeration>
599     <xs:enumeration value="se">
600         <xs:annotation>
601             <xs:documentation>Sweden</xs:documentation>
602         </xs:annotation>
603     </xs:enumeration>
604     <xs:enumeration value="si">
605         <xs:annotation>
606             <xs:documentation>Slovenia</xs:documentation>
607         </xs:annotation>
608     </xs:enumeration>
609     <xs:enumeration value="sk">
610         <xs:annotation>
611             <xs:documentation>Slovakia</xs:documentation>
612         </xs:annotation>
613     </xs:enumeration>
614     <xs:enumeration value="sm">
615         <xs:annotation>
616             <xs:documentation>San Marino</xs:documentation>
617         </xs:annotation>
618     </xs:enumeration>
619     <xs:enumeration value="tr">
620         <xs:annotation>
621             <xs:documentation>Turkey</xs:documentation>
622         </xs:annotation>
623     </xs:enumeration>
624     <xs:enumeration value="va">
625         <xs:annotation>
626             <xs:documentation>Vatican City State</xs:documentation>
627         </xs:annotation>
628     </xs:enumeration>
629     <xs:enumeration value="other">
630         <xs:annotation>
631             <xs:documentation>Other than as defined in this enumeration.</
↪ xs:documentation>
632         </xs:annotation>
633     </xs:enumeration>
634 </xs:restriction>
635 </xs:simpleType>
636 <xs:element name="d2LogicalModel" type="D2LogicalModel:D2LogicalModel">
637     <xs:unique name="_d2LogicalModelSituationRecordConstraint">
638         <xs:selector xpath="."/><xs:field xpath="situationRecord" />
639     </xs:unique>
640     <xs:unique name="_d2LogicalModelSituationConstraint">
641         <xs:selector xpath="."/><xs:field xpath="situation" />
642     </xs:unique>
643 </xs:element>

```

(continues on next page)

(continued from previous page)

```

648 <xs:complexType name="D2LogicalModel">
649   <xs:annotation>
650     <xs:documentation>The DATEX II logical model comprising exchange, content_
↪ payload and management sub-models.</xs:documentation>
651   </xs:annotation>
652   <xs:sequence>
653     <xs:element name="exchange" type="D2LogicalModel:Exchange" />
654     <xs:element name="payloadPublication" type="D2LogicalModel:PayloadPublication" _
↪ minOccurs="0" />
655     <xs:element name="d2LogicalModelExtension" type="D2LogicalModel:_ExtensionType" _
↪ minOccurs="0" />
656   </xs:sequence>
657   <xs:attribute name="modelBaseVersion" use="required" fixed="2" />
658 </xs:complexType>
659 <xs:complexType name="DataValue" abstract="true">
660   <xs:annotation>
661     <xs:documentation>A data value of something that can be measured or calculated._
↪ Any provided meta-data values specified in the attributes override any specified_
↪ generic characteristics such as defined for a specific measurement in the_
↪ MeasurementSiteTable. </xs:documentation>
662   </xs:annotation>
663   <xs:sequence>
664     <xs:element name="dataError" type="D2LogicalModel:Boolean" minOccurs="0" _
↪ maxOccurs="1">
665       <xs:annotation>
666         <xs:documentation>Indication of whether the value is deemed to be erroneous._
↪ by the supplier, (true = erroneous). If not present the data value is assumed to be_
↪ ok. This may be used when automatic fault detection information relating to sensors_
↪ is available.</xs:documentation>
667       </xs:annotation>
668     </xs:element>
669     <xs:element name="reasonForDataError" type="D2LogicalModel:MultilingualString" _
↪ minOccurs="0" maxOccurs="1">
670       <xs:annotation>
671         <xs:documentation>The reason why the value is deemed to be erroneous by the_
↪ supplier.</xs:documentation>
672       </xs:annotation>
673     </xs:element>
674     <xs:element name="dataValueExtension" type="D2LogicalModel:_ExtensionType" _
↪ minOccurs="0" />
675   </xs:sequence>
676   <xs:attribute name="accuracy" type="D2LogicalModel:Percentage" use="optional">
677     <xs:annotation>
678       <xs:documentation>The extent to which the value is expected to be free from_
↪ error, measured as a percentage of the data value. 100% means fully accurate.</_
↪ xs:documentation>
679     </xs:annotation>
680   </xs:attribute>
681   <xs:attribute name="computationalMethod" type=
↪ "D2LogicalModel:ComputationMethodEnum" use="optional">
682     <xs:annotation>
683       <xs:documentation>Method of computation which has been used to compute this_
↪ data value.</xs:documentation>
684     </xs:annotation>
685   </xs:attribute>
686   <xs:attribute name="numberOfIncompleteInputs" type=

```

(continues on next page)

(continued from previous page)

```

687     <xs:annotation>
688       <xs:documentation>The number of inputs detected but not completed during the
        ↪ sampling or measurement period; e.g. vehicles detected entering but not exiting the
        ↪ detection zone.</xs:documentation>
689     </xs:annotation>
690   </xs:attribute>
691   <xs:attribute name="numberOfInputValuesUsed" type=
        ↪ "D2LogicalModel:NonNegativeInteger" use="optional">
692     <xs:annotation>
693       <xs:documentation>The number of input values used in the sampling or
        ↪ measurment period to determine the data value.</xs:documentation>
694     </xs:annotation>
695   </xs:attribute>
696   <xs:attribute name="smoothingFactor" type="D2LogicalModel:Float" use="optional">
697     <xs:annotation>
698       <xs:documentation>Coefficient required when a moving average is computed to
        ↪ give specific weights to the former average and the new data. A typical formula is,
        ↪ F being the smoothing factor: New average = (old average) F + (new data) (1 - F).</
        ↪ xs:documentation>
699     </xs:annotation>
700   </xs:attribute>
701   <xs:attribute name="standardDeviation" type="D2LogicalModel:Float" use="optional">
702     <xs:annotation>
703       <xs:documentation>The standard deviation of the sample of input values from
        ↪ which this value was derived, measured in the units of the data value. </
        ↪ xs:documentation>
704     </xs:annotation>
705   </xs:attribute>
706   <xs:attribute name="supplierCalculatedDataQuality" type="D2LogicalModel:Percentage
        ↪ " use="optional">
707     <xs:annotation>
708       <xs:documentation>A measure of data quality assigned to the value by the
        ↪ supplier. 100% equates to ideal/perfect quality. The method of calculation is
        ↪ supplier specific and needs to be agreed between supplier and client.</
        ↪ xs:documentation>
709     </xs:annotation>
710   </xs:attribute>
711 </xs:complexType>
712 <xs:simpleType name="DateTime">
713   <xs:annotation>
714     <xs:documentation>A combination of integer-valued year, month, day, hour,
        ↪ minute properties, a decimal-valued second property and a time zone property from
        ↪ which it is possible to determine the local time, the equivalent UTC time and the
        ↪ time zone offset from UTC.</xs:documentation>
715   </xs:annotation>
716   <xs:restriction base="xs:dateTime" />
717 </xs:simpleType>
718 <xs:complexType name="DateTimeValue">
719   <xs:annotation>
720     <xs:documentation>A measured or calculated value of an instance in time.</
        ↪ xs:documentation>
721   </xs:annotation>
722   <xs:complexContent>
723     <xs:extension base="D2LogicalModel:DataValue">
724       <xs:sequence>

```

(continues on next page)

(continued from previous page)

```

725     <xs:element name="dateTime" type="D2LogicalModel:DateTime" minOccurs="1"
↪maxOccurs="1">
726         <xs:annotation>
727             <xs:documentation>A time stamp defining an instance in time.</
↪xs:documentation>
728         </xs:annotation>
729     </xs:element>
730     <xs:element name="dateTimeValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
731 </xs:sequence>
732 </xs:extension>
733 </xs:complexContent>
734 </xs:complexType>
735 <xs:complexType name="DirectionBearingValue">
736     <xs:annotation>
737         <xs:documentation>A measured or calculated value of direction as a bearing.</
↪xs:documentation>
738     </xs:annotation>
739     <xs:complexContent>
740         <xs:extension base="D2LogicalModel:DataValue">
741             <xs:sequence>
742                 <xs:element name="directionBearing" type="D2LogicalModel:AngleInDegrees"
↪minOccurs="1" maxOccurs="1">
743                     <xs:annotation>
744                         <xs:documentation>A value of direction expressed in terms of a bearing
↪measured in whole degrees. Unless otherwise specified the reference direction
↪corresponding to 0 degrees is North.</xs:documentation>
745                     </xs:annotation>
746                 </xs:element>
747                 <xs:element name="directionBearingValueExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
748             </xs:sequence>
749         </xs:extension>
750     </xs:complexContent>
751 </xs:complexType>
752 <xs:simpleType name="DirectionCompassEnum">
753     <xs:annotation>
754         <xs:documentation>Cardinal direction points of the compass.</xs:documentation>
755     </xs:annotation>
756     <xs:restriction base="xs:string">
757         <xs:enumeration value="east">
758             <xs:annotation>
759                 <xs:documentation>East.</xs:documentation>
760             </xs:annotation>
761         </xs:enumeration>
762         <xs:enumeration value="eastNorthEast">
763             <xs:annotation>
764                 <xs:documentation>East north east.</xs:documentation>
765             </xs:annotation>
766         </xs:enumeration>
767         <xs:enumeration value="eastSouthEast">
768             <xs:annotation>
769                 <xs:documentation>East south east.</xs:documentation>
770             </xs:annotation>
771         </xs:enumeration>
772         <xs:enumeration value="north">

```

(continues on next page)

(continued from previous page)

```

773     <xs:annotation>
774         <xs:documentation>North.</xs:documentation>
775     </xs:annotation>
776 </xs:enumeration>
777 <xs:enumeration value="northEast">
778     <xs:annotation>
779         <xs:documentation>North east.</xs:documentation>
780     </xs:annotation>
781 </xs:enumeration>
782 <xs:enumeration value="northNorthEast">
783     <xs:annotation>
784         <xs:documentation>North north east.</xs:documentation>
785     </xs:annotation>
786 </xs:enumeration>
787 <xs:enumeration value="northNorthWest">
788     <xs:annotation>
789         <xs:documentation>North north west.</xs:documentation>
790     </xs:annotation>
791 </xs:enumeration>
792 <xs:enumeration value="northWest">
793     <xs:annotation>
794         <xs:documentation>North west.</xs:documentation>
795     </xs:annotation>
796 </xs:enumeration>
797 <xs:enumeration value="south">
798     <xs:annotation>
799         <xs:documentation>South.</xs:documentation>
800     </xs:annotation>
801 </xs:enumeration>
802 <xs:enumeration value="southEast">
803     <xs:annotation>
804         <xs:documentation>South east.</xs:documentation>
805     </xs:annotation>
806 </xs:enumeration>
807 <xs:enumeration value="southSouthEast">
808     <xs:annotation>
809         <xs:documentation>South south east.</xs:documentation>
810     </xs:annotation>
811 </xs:enumeration>
812 <xs:enumeration value="southSouthWest">
813     <xs:annotation>
814         <xs:documentation>South south west.</xs:documentation>
815     </xs:annotation>
816 </xs:enumeration>
817 <xs:enumeration value="southWest">
818     <xs:annotation>
819         <xs:documentation>South west.</xs:documentation>
820     </xs:annotation>
821 </xs:enumeration>
822 <xs:enumeration value="west">
823     <xs:annotation>
824         <xs:documentation>West.</xs:documentation>
825     </xs:annotation>
826 </xs:enumeration>
827 <xs:enumeration value="westNorthWest">
828     <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

829     <xs:documentation>West north west.</xs:documentation>
830   </xs:annotation>
831 </xs:enumeration>
832 <xs:enumeration value="westSouthWest">
833   <xs:annotation>
834     <xs:documentation>West south west.</xs:documentation>
835   </xs:annotation>
836 </xs:enumeration>
837 </xs:restriction>
838 </xs:simpleType>
839 <xs:complexType name="DirectionCompassValue">
840   <xs:annotation>
841     <xs:documentation>A measured or calculated value of direction as a point of the
842     ↪ compass.</xs:documentation>
843   </xs:annotation>
844   <xs:complexContent>
845     <xs:extension base="D2LogicalModel:DataValue">
846       <xs:sequence>
847         <xs:element name="directionCompass" type=
848         ↪ "D2LogicalModel:DirectionCompassEnum" minOccurs="1" maxOccurs="1">
849           <xs:annotation>
850             <xs:documentation>A value of direction expressed in terms of points of
851             ↪ the compass.</xs:documentation>
852           </xs:annotation>
853         </xs:element>
854         <xs:element name="directionCompassValueExtension" type="D2LogicalModel:_
855         ↪ ExtensionType" minOccurs="0" />
856       </xs:sequence>
857     </xs:extension>
858   </xs:complexContent>
859 </xs:complexType>
860 <xs:simpleType name="DrivingConditionTypeEnum">
861   <xs:annotation>
862     <xs:documentation>Types of the perceived driving conditions.</xs:documentation>
863   </xs:annotation>
864   <xs:restriction base="xs:string">
865     <xs:enumeration value="impossible">
866       <xs:annotation>
867         <xs:documentation>Current conditions are making driving impossible.</
868         ↪ xs:documentation>
869       </xs:annotation>
870     </xs:enumeration>
871     <xs:enumeration value="hazardous">
872       <xs:annotation>
873         <xs:documentation>Driving conditions are hazardous due to environmental
874         ↪ conditions.</xs:documentation>
875       </xs:annotation>
876     </xs:enumeration>
877     <xs:enumeration value="normal">
878       <xs:annotation>
879         <xs:documentation>Driving conditions are normal.</xs:documentation>
880       </xs:annotation>
881     </xs:enumeration>
882     <xs:enumeration value="passableWithCare">
883       <xs:annotation>
884         <xs:documentation>The roadway is passable to vehicles with driver care.</

```

(continues on next page)

(continued from previous page)

```

879     </xs:annotation>
880   </xs:enumeration>
881   <xs:enumeration value="unknown">
882     <xs:annotation>
883       <xs:documentation>Driving conditions are unknown.</xs:documentation>
884     </xs:annotation>
885   </xs:enumeration>
886   <xs:enumeration value="veryHazardous">
887     <xs:annotation>
888       <xs:documentation>Driving conditions are very hazardous due to
889     <environmental conditions.</xs:documentation>
890     </xs:annotation>
891   </xs:enumeration>
892   <xs:enumeration value="winterConditions">
893     <xs:annotation>
894       <xs:documentation>Driving conditions are consistent with those expected in
895     <winter.</xs:documentation>
896     </xs:annotation>
897   </xs:enumeration>
898   <xs:enumeration value="other">
899     <xs:annotation>
900       <xs:documentation>Other than as defined in this enumeration.</
901     <xs:documentation>
902   </xs:annotation>
903 </xs:enumeration>
904 </xs:restriction>
905 </xs:simpleType>
906 <xs:complexType name="ElaboratedDataFault">
907   <xs:annotation>
908     <xs:documentation>Details of a fault which is being reported for the related
909   <elaborated data.</xs:documentation>
910   </xs:annotation>
911   <xs:complexContent>
912     <xs:extension base="D2LogicalModel:Fault">
913       <xs:sequence>
914         <xs:element name="elaboratedDataFault" type=
915       <"D2LogicalModel:ElaboratedDataFaultEnum" minOccurs="1" maxOccurs="1">
916         <xs:annotation>
917           <xs:documentation>The type of fault which is being reported for the
918       <specified elaborated data.</xs:documentation>
919         </xs:annotation>
920       </xs:element>
921       <xs:element name="elaboratedDataFaultExtension" type="D2LogicalModel:_
922     <ExtensionType" minOccurs="0" />
923       </xs:sequence>
924     </xs:extension>
925   </xs:complexContent>
926 </xs:complexType>
927 <xs:simpleType name="ElaboratedDataFaultEnum">
928   <xs:annotation>
929     <xs:documentation>Types of elaborated data faults.</xs:documentation>
930   </xs:annotation>
931   <xs:restriction base="xs:string">
932     <xs:enumeration value="intermittentDataValues">
933       <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

927     <xs:documentation>Data values are being produced at intermittent intervals.
↪which are not consistent with the expected reporting interval.</xs:documentation>
928     </xs:annotation>
929   </xs:enumeration>
930   <xs:enumeration value="noDataValuesAvailable">
931     <xs:annotation>
932       <xs:documentation>No elaborated data values are currently available.</
↪xs:documentation>
933     </xs:annotation>
934   </xs:enumeration>
935   <xs:enumeration value="spuriousUnreliableDataValues">
936     <xs:annotation>
937       <xs:documentation>Spurious or unreliable data values are being produced.</
↪xs:documentation>
938     </xs:annotation>
939   </xs:enumeration>
940   <xs:enumeration value="unspecifiedOrUnknownFault">
941     <xs:annotation>
942       <xs:documentation>An unspecified or unknown fault exists in the process.
↪which is generating elaborated data.</xs:documentation>
943     </xs:annotation>
944   </xs:enumeration>
945   <xs:enumeration value="other">
946     <xs:annotation>
947       <xs:documentation>Other than as defined in this enumeration.</
↪xs:documentation>
948     </xs:annotation>
949   </xs:enumeration>
950 </xs:restriction>
951 </xs:simpleType>
952 <xs:complexType name="Exchange">
953   <xs:annotation>
954     <xs:documentation>Details associated with the management of the exchange.
↪between the supplier and the client.</xs:documentation>
955   </xs:annotation>
956   <xs:sequence>
957     <xs:element name="supplierIdentification" type=
↪"D2LogicalModel:InternationalIdentifier" />
958     <xs:element name="exchangeExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
959   </xs:sequence>
960 </xs:complexType>
961 <xs:complexType name="Fault">
962   <xs:annotation>
963     <xs:documentation>Information about a fault relating to a specific piece of
↪equipment or process.</xs:documentation>
964   </xs:annotation>
965   <xs:sequence>
966     <xs:element name="faultIdentifier" type="D2LogicalModel:String" minOccurs="0"
↪maxOccurs="1">
967       <xs:annotation>
968         <xs:documentation>Unique identifier of the fault.</xs:documentation>
969       </xs:annotation>
970     </xs:element>
971     <xs:element name="faultDescription" type="D2LogicalModel:String" minOccurs="0"
↪maxOccurs="1">

```

(continues on next page)

(continued from previous page)

```

972     <xs:annotation>
973       <xs:documentation>Textual description of the fault.</xs:documentation>
974     </xs:annotation>
975   </xs:element>
976   <xs:element name="faultCreationTime" type="D2LogicalModel:DateTime" minOccurs="0
↪ " maxOccurs="1">
977     <xs:annotation>
978       <xs:documentation>The date and time at which the fault was originally_
↪ recorded/reported.</xs:documentation>
979     </xs:annotation>
980   </xs:element>
981   <xs:element name="faultLastUpdateTime" type="D2LogicalModel:DateTime" minOccurs=
↪ "1" maxOccurs="1">
982     <xs:annotation>
983       <xs:documentation>The date and time at which the fault information as_
↪ specified in this instance was last updated.</xs:documentation>
984     </xs:annotation>
985   </xs:element>
986   <xs:element name="faultSeverity" type="D2LogicalModel:FaultSeverityEnum"
↪ minOccurs="0" maxOccurs="1">
987     <xs:annotation>
988       <xs:documentation>The severity of the fault in terms of how it affects the_
↪ usability of the equipment or the reliability of the data generated by the_
↪ equipment.</xs:documentation>
989     </xs:annotation>
990   </xs:element>
991   <xs:element name="faultExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
992 </xs:sequence>
993 </xs:complexType>
994 <xs:simpleType name="FaultSeverityEnum">
995   <xs:annotation>
996     <xs:documentation>Classification of the severity of faults.</xs:documentation>
997   </xs:annotation>
998   <xs:restriction base="xs:string">
999     <xs:enumeration value="low">
1000       <xs:annotation>

```

```

1001     <xs:documentation>The fault is of low severity and has only limited impact_
↪ on the usability of the equipment or the value of the data generated by the_
↪ equipment.</xs:documentation>
1002   </xs:annotation>
1003 </xs:enumeration>
1004   <xs:enumeration value="medium">
1005     <xs:annotation>
1006       <xs:documentation>The fault is of medium severity which will significantly_
↪ limit the usability of the equipment or devalue the usefulness of the data_
↪ generated by the equipment.</xs:documentation>
1007     </xs:annotation>
1008 </xs:enumeration>
1009   <xs:enumeration value="high">
1010     <xs:annotation>
1011       <xs:documentation>The fault is of high severity which will render the_
↪ equipment unusable or any data generated by the equipment to be of no value.</
↪ xs:documentation>
1012     </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

1013     </xs:enumeration>
1014     <xs:enumeration value="unknown">
1015         <xs:annotation>
1016             <xs:documentation>The fault is of unknown severity and hence its effect on
↪ the usability of the equipment or the usefulness of the data generated by the
↪ equipment can not be assessed.</xs:documentation>
1017         </xs:annotation>
1018     </xs:enumeration>
1019 </xs:restriction>
1020 </xs:simpleType>
1021 <xs:simpleType name="Float">
1022     <xs:annotation>
1023         <xs:documentation>A floating point number whose value space consists of the
↪ values  $m \times 2^e$ , where  $m$  is an integer whose absolute value is less than  $2^{24}$ , and  $e$ 
↪ is an integer between -149 and 104, inclusive.</xs:documentation>
1024     </xs:annotation>
1025     <xs:restriction base="xs:float" />
1026 </xs:simpleType>
1027 <xs:complexType name="FloatingPointMetreDistanceValue">
1028     <xs:annotation>
1029         <xs:documentation>A measured or calculated value of distance in metres in a
↪ floating point format.</xs:documentation>
1030     </xs:annotation>
1031     <xs:complexContent>
1032         <xs:extension base="D2LogicalModel:DataValue">
1033             <xs:sequence>
1034                 <xs:element name="floatingPointMetreDistance" type=
↪ "D2LogicalModel:MetresAsFloat" minOccurs="1" maxOccurs="1">
1035                     <xs:annotation>
1036                         <xs:documentation>A value of distance expressed in metres in a floating
↪ point format.</xs:documentation>
1037                     </xs:annotation>
1038                 </xs:element>
1039                 <xs:element name="floatingPointMetreDistanceValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1040             </xs:sequence>
1041         </xs:extension>
1042     </xs:complexContent>
1043 </xs:complexType>
1044 <xs:complexType name="GroupOfLocations" abstract="true">
1045     <xs:annotation>
1046         <xs:documentation>One or more physically separate locations. Multiple locations
↪ may be related, as in an itinerary (or route), or may be unrelated. It is not for
↪ identifying the same physical location using different Location objects for
↪ different referencing systems.</xs:documentation>
1047     </xs:annotation>
1048     <xs:sequence>
1049         <xs:element name="groupOfLocationsExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
1050     </xs:sequence>
1051 </xs:complexType>
1052 <xs:complexType name="HeaderInformation">
1053     <xs:annotation>
1054         <xs:documentation>Management information relating to the data contained within
↪ a publication.</xs:documentation>
1055     </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

1056     <xs:sequence>
1057       <xs:element name="confidentiality" type="D2LogicalModel:ConfidentialityValueEnum
↪ " minOccurs="1" maxOccurs="1">
1058         <xs:annotation>
1059           <xs:documentation>The extent to which the related information may be
↪ circulated, according to the recipient type. Recipients must comply with this
↪ confidentiality statement.</xs:documentation>
1060         </xs:annotation>
1061       </xs:element>
1062       <xs:element name="informationStatus" type="D2LogicalModel:InformationStatusEnum
↪ " minOccurs="1" maxOccurs="1">
1063         <xs:annotation>
1064           <xs:documentation>The status of the related information (real, test,
↪ exercise ....).</xs:documentation>
1065         </xs:annotation>
1066       </xs:element>
1067       <xs:element name="urgency" type="D2LogicalModel:UrgencyEnum" minOccurs="0"
↪ maxOccurs="1">
1068         <xs:annotation>
1069           <xs:documentation>This indicates the urgency with which a message recipient
↪ or Client should distribute the enclosed information. Urgency particularly relates
↪ to functions within RDS-TMC applications. </xs:documentation>
1070         </xs:annotation>
1071       </xs:element>
1072       <xs:element name="headerInformationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1073     </xs:sequence>
1074   </xs:complexType>
1075   <xs:simpleType name="InformationStatusEnum">
1076     <xs:annotation>
1077       <xs:documentation>Status of the related information (i.e. real, test or
↪ exercise).</xs:documentation>
1078     </xs:annotation>
1079     <xs:restriction base="xs:string">
1080       <xs:enumeration value="real">
1081         <xs:annotation>
1082           <xs:documentation>The information is real. It is not a test or exercise.</
↪ xs:documentation>
1083         </xs:annotation>
1084       </xs:enumeration>
1085       <xs:enumeration value="securityExercise">
1086         <xs:annotation>
1087           <xs:documentation>The information is part of an exercise which is for
↪ testing security.</xs:documentation>
1088         </xs:annotation>
1089       </xs:enumeration>
1090       <xs:enumeration value="technicalExercise">
1091         <xs:annotation>
1092           <xs:documentation>The information is part of an exercise which includes
↪ tests of associated technical subsystems.</xs:documentation>
1093         </xs:annotation>
1094       </xs:enumeration>
1095       <xs:enumeration value="test">
1096         <xs:annotation>
1097           <xs:documentation>The information is part of a test for checking the
↪ exchange of this type of information.</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

1098     </xs:annotation>
1099     </xs:enumeration>
1100   </xs:restriction>
1101 </xs:simpleType>
1102 <xs:simpleType name="Integer">
1103   <xs:annotation>
1104     <xs:documentation>An integer number whose value space is the set {-2147483648, -
↪ 2147483647, -2147483646, ..., -2, -1, 0, 1, 2, ..., 2147483645, 2147483646,
↪ 2147483647}.</xs:documentation>
1105   </xs:annotation>
1106   <xs:restriction base="xs:integer" />
1107 </xs:simpleType>
1108 <xs:complexType name="IntegerMetreDistanceValue">
1109   <xs:annotation>
1110     <xs:documentation>A measured or calculated value of distance in whole metres.</
↪ xs:documentation>
1111   </xs:annotation>
1112   <xs:complexContent>
1113     <xs:extension base="D2LogicalModel:DataValue">
1114       <xs:sequence>
1115         <xs:element name="integerMetreDistance" type=
↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="1" maxOccurs="1">
1116           <xs:annotation>
1117             <xs:documentation>A value of distance expressed in metres in a non-
↪ negative integer format.</xs:documentation>
1118           </xs:annotation>
1119         </xs:element>
1120         <xs:element name="integerMetreDistanceValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1121       </xs:sequence>
1122     </xs:extension>
1123   </xs:complexContent>
1124 </xs:complexType>
1125 <xs:simpleType name="IntensityKilogramsPerSquareMetre">
1126   <xs:annotation>
1127     <xs:documentation>A measure of the quantity of application of a substance to an
↪ area defined in kilograms per square metre.</xs:documentation>
1128   </xs:annotation>
1129   <xs:restriction base="D2LogicalModel:Float" />
1130 </xs:simpleType>
1131 <xs:simpleType name="IntensityMillimetresPerHour">
1132   <xs:annotation>
1133     <xs:documentation>A measure of precipitation intensity defined in millimetres
↪ per hour.</xs:documentation>
1134   </xs:annotation>
1135   <xs:restriction base="D2LogicalModel:Float" />
1136 </xs:simpleType>
1137 <xs:complexType name="InternationalIdentifier">
1138   <xs:annotation>
1139     <xs:documentation>An identifier/name whose range is specific to the particular
↪ country.</xs:documentation>
1140   </xs:annotation>
1141   <xs:sequence>
1142     <xs:element name="country" type="D2LogicalModel:CountryEnum" minOccurs="1"
↪ maxOccurs="1">
1143       <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

1144     <xs:documentation>ISO 3166-1 two character country code.</xs:documentation>
1145     </xs:annotation>
1146   </xs:element>
1147   <xs:element name="nationalIdentifier" type="D2LogicalModel:String" minOccurs="1
1148   ↪ " maxOccurs="1">
1149     <xs:annotation>
1150       <xs:documentation>Identifier or name unique within the specified country.</
1151       ↪ xs:documentation>
1152     </xs:annotation>
1153     </xs:element>
1154     <xs:element name="internationalIdentifierExtension" type="D2LogicalModel:_
1155     ↪ ExtensionType" minOccurs="0" />
1156   </xs:sequence>
1157 </xs:complexType>
1158 <xs:complexType name="KilogramsConcentrationValue">
1159   <xs:annotation>
1160     <xs:documentation>A measured or calculated value of concentration of a
1161     ↪ substance in grams per unit volume.</xs:documentation>
1162   </xs:annotation>
1163   <xs:complexContent>
1164     <xs:extension base="D2LogicalModel:DataValue">
1165       <xs:sequence>
1166         <xs:element name="kilogramsConcentration" type=
1167         ↪ "D2LogicalModel:ConcentrationKilogramsPerCubicMetre" minOccurs="1" maxOccurs="1">
1168           <xs:annotation>
1169             <xs:documentation>A value defining the amount of a substance in a given
1170             ↪ volume (concentration) expressed in kilograms per cubic metre.</xs:documentation>
1171           </xs:annotation>
1172           </xs:element>
1173           <xs:element name="kilogramsConcentrationValueExtension" type=
1174           ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1175         </xs:sequence>
1176       </xs:extension>
1177     </xs:complexContent>
1178   </xs:complexType>
1179   <xs:simpleType name="KilometresPerHour">
1180     <xs:annotation>
1181       <xs:documentation>A measure of speed defined in kilometres per hour.</
1182       ↪ xs:documentation>
1183     </xs:annotation>
1184     <xs:restriction base="D2LogicalModel:Float" />
1185   </xs:simpleType>
1186   <xs:simpleType name="Language">
1187     <xs:annotation>
1188       <xs:documentation>A language datatype, identifies a specified language by an
1189       ↪ ISO 639-1 2-alpha / ISO 639-2 3-alpha code.</xs:documentation>
1190     </xs:annotation>
1191     <xs:restriction base="xs:language" />
1192   </xs:simpleType>
1193   <xs:complexType name="Location" abstract="true">
1194     <xs:annotation>
1195       <xs:documentation>The specification of a location either on a network (as a
1196       ↪ point or a linear location) or as an area. This may be provided in one or more
1197       ↪ referencing systems.</xs:documentation>
1198     </xs:annotation>
1199   </xs:complexType>

```

(continues on next page)

(continued from previous page)

```

1189     <xs:extension base="D2LogicalModel:GroupOfLocations">
1190         <xs:sequence>
1191             <xs:element name="locationExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
1192         </xs:sequence>
1193     </xs:extension>
1194 </xs:complexContent>
1195 </xs:complexType>
1196 <xs:complexType name="LocationByReference">
1197     <xs:annotation>
1198         <xs:documentation>A location defined by reference to a predefined location.</
↪ xs:documentation>
1199     </xs:annotation>
1200     <xs:complexContent>
1201         <xs:extension base="D2LogicalModel:Location">
1202             <xs:sequence>
1203                 <xs:element name="predefinedLocationReference" type="D2LogicalModel:_
↪ PredefinedLocationVersionedReference" minOccurs="1" maxOccurs="1">
1204                     <xs:annotation>
1205                         <xs:documentation>A reference to a versioned predefined location.</
↪ xs:documentation>
1206                     </xs:annotation>
1207                 </xs:element>
1208                 <xs:element name="locationByReferenceExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1209             </xs:sequence>
1210         </xs:extension>
1211     </xs:complexContent>
1212 </xs:complexType>
1213 <xs:complexType name="MeasurementEquipmentFault">
1214     <xs:annotation>
1215         <xs:documentation>Details of a fault which is being reported for the related
↪ measurement equipment.</xs:documentation>
1216     </xs:annotation>
1217     <xs:complexContent>
1218         <xs:extension base="D2LogicalModel:Fault">
1219             <xs:sequence>
1220                 <xs:element name="measurementEquipmentFault" type=
↪ "D2LogicalModel:MeasurementEquipmentFaultEnum" minOccurs="1" maxOccurs="1">
1221                     <xs:annotation>
1222                         <xs:documentation>The type of fault which is being reported for the
↪ specified measurement equipment.</xs:documentation>
1223                     </xs:annotation>
1224                 </xs:element>
1225                 <xs:element name="measurementEquipmentFaultExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1226             </xs:sequence>
1227         </xs:extension>
1228     </xs:complexContent>
1229 </xs:complexType>
1230 <xs:simpleType name="MeasurementEquipmentFaultEnum">
1231     <xs:annotation>
1232         <xs:documentation>Types of measurement equipment faults.</xs:documentation>
1233     </xs:annotation>
1234     <xs:restriction base="xs:string">
1235         <xs:enumeration value="intermittentDataValues">

```

(continues on next page)

(continued from previous page)

```

1236     <xs:annotation>
1237       <xs:documentation>Data values are being produced at intermittent intervals.
↪ which are not consistent with the expected reporting interval.</xs:documentation>
1238     </xs:annotation>
1239   </xs:enumeration>
1240   <xs:enumeration value="noDataValuesAvailable">
1241     <xs:annotation>
1242       <xs:documentation>No measured data values are currently available.</
↪ xs:documentation>
1243     </xs:annotation>
1244   </xs:enumeration>
1245   <xs:enumeration value="spuriousUnreliableDataValues">
1246     <xs:annotation>
1247       <xs:documentation>Spurious or unreliable data values are being produced.</
↪ xs:documentation>
1248     </xs:annotation>
1249   </xs:enumeration>
1250   <xs:enumeration value="unspecifiedOrUnknownFault">
1251     <xs:annotation>
1252       <xs:documentation>An unspecified or unknown fault exists in the measurement.
↪ equipment.</xs:documentation>
1253     </xs:annotation>
1254   </xs:enumeration>
1255   <xs:enumeration value="other">
1256     <xs:annotation>
1257       <xs:documentation>Other than as defined in this enumeration.</
↪ xs:documentation>
1258     </xs:annotation>
1259   </xs:enumeration>
1260 </xs:restriction>
1261 </xs:simpleType>
1262 <xs:simpleType name="MetresAsFloat">
1263   <xs:annotation>
1264     <xs:documentation>A measure of distance defined in metres in a floating point.
↪ format.</xs:documentation>
1265   </xs:annotation>
1266   <xs:restriction base="D2LogicalModel:Float" />
1267 </xs:simpleType>
1268 <xs:simpleType name="MetresAsNonNegativeInteger">
1269   <xs:annotation>
1270     <xs:documentation>A measure of distance defined in metres in a non negative.
↪ integer format.</xs:documentation>
1271   </xs:annotation>
1272   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1273 </xs:simpleType>
1274 <xs:complexType name="MultilingualString">
1275   <xs:sequence>
1276     <xs:element name="values">
1277       <xs:complexType>
1278         <xs:sequence>
1279           <xs:element name="value" type="D2LogicalModel:MultilingualStringValue"
↪ maxOccurs="unbounded" />
1280         </xs:sequence>
1281       </xs:complexType>
1282     </xs:element>
1283   </xs:sequence>

```

(continues on next page)

(continued from previous page)

```

1284 </xs:complexType>
1285 <xs:complexType name="MultilingualStringValue">
1286   <xs:simpleContent>
1287     <xs:extension base="D2LogicalModel:MultilingualStringValue" type="D2LogicalModel:MultilingualStringValue" />
1288     <xs:attribute name="lang" type="xs:language" />
1289   </xs:simpleContent>
1290 </xs:complexType>
1291 <xs:simpleType name="MultilingualStringValue" type="xs:string">
1292   <xs:restriction base="xs:string">
1293     <xs:maxLength value="1024" />
1294   </xs:restriction>
1295 </xs:simpleType>
1296 <xs:complexType name="NonManagedCause">
1297   <xs:annotation>
1298     <xs:documentation>A cause of this situation record which is not managed by the
1299     ↪ publication creator, i.e. one which is not represented by another situation record
1300     ↪ produced by the same publication creator.</xs:documentation>
1301   </xs:annotation>
1302   <xs:complexContent>
1303     <xs:extension base="D2LogicalModel:Cause">
1304       <xs:sequence>
1305         <xs:element name="causeType" type="D2LogicalModel:CauseTypeEnum" minOccurs="0" maxOccurs="1">
1306           <xs:annotation>
1307             <xs:documentation>Indicates an external influence that may be the
1308             ↪ causation of components of a situation.</xs:documentation>
1309           </xs:annotation>
1310         </xs:element>
1311         <xs:element name="nonManagedCauseExtension" type="D2LogicalModel:_
1312         ↪ ExtensionType" minOccurs="0" />
1313       </xs:sequence>
1314     </xs:extension>
1315   </xs:complexContent>
1316 </xs:complexType>
1317 <xs:simpleType name="NonNegativeInteger">
1318   <xs:annotation>
1319     <xs:documentation>An integer number whose value space is the set {0, 1, 2, ...,
1320     ↪ 2147483645, 2147483646, 2147483647}.</xs:documentation>
1321   </xs:annotation>
1322   <xs:restriction base="xs:nonNegativeInteger" />
1323 </xs:simpleType>
1324 <xs:complexType name="NonOrderedLocationGroupByReference">
1325   <xs:annotation>
1326     <xs:documentation>A group of (i.e. more than one) physically separate locations
1327     ↪ which have no specific order that are defined by reference to a predefined non-
1328     ↪ ordered location group.</xs:documentation>
1329   </xs:annotation>
1330   <xs:complexContent>
1331     <xs:extension base="D2LogicalModel:NonOrderedLocations">
1332       <xs:sequence>
1333         <xs:element name="predefinedNonOrderedLocationGroupReference" type="
1334         ↪ "D2LogicalModel:_PredefinedNonOrderedLocationGroupVersionedReference" minOccurs="1"
1335         ↪ maxOccurs="1">
1336           <xs:annotation>
1337             <xs:documentation>A reference to a versioned instance of a predefined

```

(continues on next page)

(continued from previous page)

```

↪non ordered location group as specified in a PredefinedLocationsPublication.</
↪xs:documentation>
1330     </xs:annotation>
1331     </xs:element>
1332     <xs:element name="nonOrderedLocationGroupByReferenceExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
1333     </xs:sequence>
1334     </xs:extension>
1335     </xs:complexContent>
1336 </xs:complexType>
1337 <xs:complexType name="NonOrderedLocations" abstract="true">
1338     <xs:annotation>
1339     <xs:documentation>Multiple (i.e. more than one) physically separate locations,
↪which have no specific order.</xs:documentation>
1340     </xs:annotation>
1341     <xs:complexContent>
1342     <xs:extension base="D2LogicalModel:GroupOfLocations">
1343     <xs:sequence>
1344     <xs:element name="nonOrderedLocationsExtension" type="D2LogicalModel:_
↪ExtensionType" minOccurs="0" />
1345     </xs:sequence>
1346     </xs:extension>
1347     </xs:complexContent>
1348 </xs:complexType>
1349 <xs:complexType name="NonWeatherRelatedRoadConditions">
1350     <xs:annotation>
1351     <xs:documentation>Road surface conditions that are not related to the weather,
↪but which may affect driving conditions.</xs:documentation>
1352     </xs:annotation>
1353     <xs:complexContent>
1354     <xs:extension base="D2LogicalModel:RoadConditions">
1355     <xs:sequence>
1356     <xs:element name="nonWeatherRelatedRoadConditionType" type=
↪"D2LogicalModel:NonWeatherRelatedRoadConditionTypeEnum" minOccurs="1" maxOccurs=
↪"unbounded">
1357     <xs:annotation>
1358     <xs:documentation>The type of road conditions which are not related to,
↪the weather.</xs:documentation>
1359     </xs:annotation>
1360     </xs:element>
1361     <xs:element name="nonWeatherRelatedRoadConditionsExtension" type=
↪"D2LogicalModel:_ExtensionType" minOccurs="0" />
1362     </xs:sequence>
1363     </xs:extension>
1364     </xs:complexContent>
1365 </xs:complexType>
1366 <xs:simpleType name="NonWeatherRelatedRoadConditionTypeEnum">
1367     <xs:annotation>
1368     <xs:documentation>Types of road surface conditions which are not related to the,
↪weather.</xs:documentation>
1369     </xs:annotation>
1370     <xs:restriction base="xs:string">
1371     <xs:enumeration value="dieselOnRoad">
1372     <xs:annotation>
1373     <xs:documentation>Increased skid risk due to diesel on the road.</
↪xs:documentation>

```

(continues on next page)

(continued from previous page)

```

1374     </xs:annotation>
1375   </xs:enumeration>
1376   <xs:enumeration value="leavesOnRoad">
1377     <xs:annotation>
1378       <xs:documentation>Increased skid risk due to leaves on road.</
↪xs:documentation>
1379     </xs:annotation>
1380   </xs:enumeration>
1381   <xs:enumeration value="looseChippings">
1382     <xs:annotation>
1383       <xs:documentation>Increased skid risk and injury risk due to loose_
↪chippings on road.</xs:documentation>
1384     </xs:annotation>
1385   </xs:enumeration>
1386   <xs:enumeration value="looseSandOnRoad">
1387     <xs:annotation>
1388       <xs:documentation>Increased skid risk due to loose sand on road.</
↪xs:documentation>
1389     </xs:annotation>
1390   </xs:enumeration>
1391   <xs:enumeration value="mudOnRoad">
1392     <xs:annotation>
1393       <xs:documentation>Increased skid risk due to mud on road.</xs:documentation>
1394     </xs:annotation>
1395   </xs:enumeration>
1396   <xs:enumeration value="oilOnRoad">
1397     <xs:annotation>
1398       <xs:documentation>Increased skid risk due to oil on road.</xs:documentation>
1399     </xs:annotation>
1400   </xs:enumeration>
1401   <xs:enumeration value="petrolOnRoad">
1402     <xs:annotation>
1403       <xs:documentation>Increased skid risk due to petrol on road.</
↪xs:documentation>
1404     </xs:annotation>
1405   </xs:enumeration>
1406   <xs:enumeration value="roadSurfaceInPoorCondition">
1407     <xs:annotation>
1408       <xs:documentation>The road surface is damaged, severely rutted or potholed_
↪(i.e. it is in a poor state of repair).</xs:documentation>
1409     </xs:annotation>
1410   </xs:enumeration>
1411   <xs:enumeration value="slipperyRoad">
1412     <xs:annotation>
1413       <xs:documentation>The road surface is slippery due to an unspecified non-
↪weather related cause.</xs:documentation>
1414     </xs:annotation>
1415   </xs:enumeration>
1416   <xs:enumeration value="other">
1417     <xs:annotation>
1418       <xs:documentation>Other than as defined in this enumeration.</
↪xs:documentation>
1419     </xs:annotation>
1420   </xs:enumeration>
1421 </xs:restriction>
1422 </xs:simpleType>

```

(continues on next page)

(continued from previous page)

```

1423 <xs:complexType name="OccupancyChangeValue">
1424   <xs:annotation>
1425     <xs:documentation>A measured or calculated value of change of occupied parking_
↪ spaces expressed as integer.</xs:documentation>
1426   </xs:annotation>
1427   <xs:complexContent>
1428     <xs:extension base="D2LogicalModel:DataValue">
1429       <xs:sequence>
1430         <xs:element name="occupancyChange" type="D2LogicalModel:Integer" minOccurs=
↪ "1" maxOccurs="1">
1431           <xs:annotation>
1432             <xs:documentation>A measured or calculated absolut change of occupied_
↪ parking spaces within a specified time expressed as integer.</xs:documentation>
1433           </xs:annotation>
1434         </xs:element>
1435         <xs:element name="occupancyChangeValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1436       </xs:sequence>
1437     </xs:extension>
1438   </xs:complexContent>
1439 </xs:complexType>
1440 <xs:complexType name="OverallPeriod">
1441   <xs:annotation>
1442     <xs:documentation>A continuous or discontinuous period of validity defined by_
↪ overall bounding start and end times and the possible intersection of valid periods_
↪ (potentially recurring) with the complement of exception periods (also potentially_
↪ recurring).</xs:documentation>
1443   </xs:annotation>
1444   <xs:sequence>
1445     <xs:element name="overallStartTime" type="D2LogicalModel:DateTime" minOccurs="1
↪ " maxOccurs="1">
1446       <xs:annotation>
1447         <xs:documentation>Start of bounding period of validity defined by date and_
↪ time.</xs:documentation>
1448       </xs:annotation>
1449     </xs:element>
1450     <xs:element name="overallEndTime" type="D2LogicalModel:DateTime" minOccurs="0"
↪ maxOccurs="1">
1451       <xs:annotation>
1452         <xs:documentation>End of bounding period of validity defined by date and_
↪ time.</xs:documentation>
1453       </xs:annotation>
1454     </xs:element>
1455     <xs:element name="overallPeriodExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
1456   </xs:sequence>
1457 </xs:complexType>
1458 <xs:simpleType name="PassengerCarUnitsPerHour">
1459   <xs:annotation>
1460     <xs:documentation>Passenger car units per hour.</xs:documentation>
1461   </xs:annotation>
1462   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
1463 </xs:simpleType>
1464 <xs:complexType name="PayloadPublication" abstract="true">
1465   <xs:annotation>
1466     <xs:documentation>A payload publication of traffic related information or_

```

(continues on next page)

(continued from previous page)

```

↪ associated management information created at a specific point in time that can be.
↪ exchanged via a DATEX II interface.</xs:documentation>
1467   </xs:annotation>
1468   <xs:sequence>
1469     <xs:element name="feedType" type="D2LogicalModel:String" minOccurs="0"
↪ maxOccurs="1">
1470       <xs:annotation>
1471         <xs:documentation>A classification of the information which is to be found.
↪ in the publications originating from the particular feed.</xs:documentation>
1472       </xs:annotation>
1473     </xs:element>
1474     <xs:element name="publicationTime" type="D2LogicalModel:DateTime" minOccurs="1"
↪ maxOccurs="1">
1475       <xs:annotation>
1476         <xs:documentation>Date/time at which the payload publication was created.</
↪ xs:documentation>
1477       </xs:annotation>
1478     </xs:element>
1479     <xs:element name="publicationCreator" type=
↪ "D2LogicalModel:InternationalIdentifier" />
1480     <xs:element name="payloadPublicationExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1481   </xs:sequence>
1482   <xs:attribute name="lang" type="D2LogicalModel:Language" use="required">
1483     <xs:annotation>
1484       <xs:documentation>The default language used throughout the payload.
↪ publication.</xs:documentation>
1485     </xs:annotation>
1486   </xs:attribute>
1487 </xs:complexType>
1488 <xs:complexType name="PcuFlowValue">
1489   <xs:annotation>
1490     <xs:documentation>A measured or calculated value of the flow rate of passenger.
↪ car units.</xs:documentation>
1491   </xs:annotation>
1492   <xs:complexContent>
1493     <xs:extension base="D2LogicalModel:DataValue">
1494       <xs:sequence>
1495         <xs:element name="pcuFlowRate" type="D2LogicalModel:PassengerCarUnitsPerHour
↪ " minOccurs="1" maxOccurs="1">
1496           <xs:annotation>
1497             <xs:documentation>A value of passenger car unit flow rate expressed in.
↪ passenger car units per hour.</xs:documentation>
1498           </xs:annotation>
1499         </xs:element>
1500         <xs:element name="pcuFlowValueExtension" type="D2LogicalModel:_ExtensionType
↪ " minOccurs="0" />
1501       </xs:sequence>
1502     </xs:extension>
1503   </xs:complexContent>
1504 </xs:complexType>
1505 <xs:simpleType name="Percentage">
1506   <xs:annotation>
1507     <xs:documentation>A measure of percentage.</xs:documentation>
1508   </xs:annotation>
1509   <xs:restriction base="D2LogicalModel:Float" />

```

(continues on next page)

(continued from previous page)

```

1510 </xs:simpleType>
1511 <xs:complexType name="PoorEnvironmentConditions">
1512   <xs:annotation>
1513     <xs:documentation>Any environmental conditions which may be affecting the_
↪ driving conditions on the road.</xs:documentation>
1514   </xs:annotation>
1515   <xs:complexContent>
1516     <xs:extension base="D2LogicalModel:Conditions">
1517       <xs:sequence>
1518         <xs:element name="poorEnvironmentType" type=
↪ "D2LogicalModel:PoorEnvironmentTypeEnum" minOccurs="1" maxOccurs="unbounded">
1519           <xs:annotation>
1520             <xs:documentation>The type of environment condition which is affecting_
↪ driving conditions.</xs:documentation>
1521           </xs:annotation>
1522           </xs:element>
1523           <xs:element name="precipitationDetail" type=
↪ "D2LogicalModel:PrecipitationDetail" minOccurs="0" />
1524           <xs:element name="visibility" type="D2LogicalModel:Visibility" minOccurs="0"
↪ " />
1525           <xs:element name="temperature" type="D2LogicalModel:Temperature" minOccurs=
↪ "0" />
1526           <xs:element name="wind" type="D2LogicalModel:Wind" minOccurs="0" />
1527           <xs:element name="poorEnvironmentConditionsExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1528         </xs:sequence>
1529       </xs:extension>
1530     </xs:complexContent>
1531   </xs:complexType>
1532   <xs:simpleType name="PoorEnvironmentTypeEnum">
1533     <xs:annotation>
1534       <xs:documentation>Types of poor environmental conditions.</xs:documentation>
1535     </xs:annotation>
1536     <xs:restriction base="xs:string">
1537       <xs:enumeration value="badWeather">
1538         <xs:annotation>
1539           <xs:documentation>Adverse weather conditions are affecting driving_
↪ conditions.</xs:documentation>
1540         </xs:annotation>
1541       </xs:enumeration>
1542       <xs:enumeration value="blizzard">
1543         <xs:annotation>
1544           <xs:documentation>Heavy snowfall in combination with strong winds, limiting_
↪ visibility to 50m or less.</xs:documentation>
1545         </xs:annotation>
1546       </xs:enumeration>
1547       <xs:enumeration value="blowingDust">
1548         <xs:annotation>
1549           <xs:documentation>Dust blowing across the roadway causing significantly_
↪ reduced visibility.</xs:documentation>
1550         </xs:annotation>
1551       </xs:enumeration>
1552       <xs:enumeration value="blowingSnow">
1553         <xs:annotation>
1554           <xs:documentation>Fallen snow moving due to the forces of wind.</
↪ xs:documentation>

```

(continues on next page)

(continued from previous page)

```

1555     </xs:annotation>
1556   </xs:enumeration>
1557   <xs:enumeration value="crosswinds">
1558     <xs:annotation>
1559       <xs:documentation>Strong cross winds across the direction of the roadway (e.
↪g. on a ridge or bridge).</xs:documentation>
1560     </xs:annotation>
1561   </xs:enumeration>
1562   <xs:enumeration value="damagingHail">
1563     <xs:annotation>
1564       <xs:documentation>Large falling ice pellets or frozen rain capable of
↪causing injury or damage to property.</xs:documentation>
1565     </xs:annotation>
1566   </xs:enumeration>
1567   <xs:enumeration value="denseFog">
1568     <xs:annotation>
1569       <xs:documentation>Dense fog, limiting visibility to 50m or less.</
↪xs:documentation>
1570     </xs:annotation>
1571   </xs:enumeration>
1572   <xs:enumeration value="eclipse">
1573     <xs:annotation>
1574       <xs:documentation>Eclipse, either partial or full, of the sun causing low
↪light levels during normal daylight period.</xs:documentation>
1575     </xs:annotation>
1576   </xs:enumeration>
1577   <xs:enumeration value="extremeCold">
1578     <xs:annotation>
1579       <xs:documentation>Abnormally low temperatures.</xs:documentation>
1580     </xs:annotation>
1581   </xs:enumeration>
1582   <xs:enumeration value="extremeHeat">
1583     <xs:annotation>
1584       <xs:documentation>Abnormally high expected maximum temperature.</
↪xs:documentation>
1585     </xs:annotation>
1586   </xs:enumeration>
1587   <xs:enumeration value="fog">
1588     <xs:annotation>
1589       <xs:documentation>Fog, visibility more than 50m.</xs:documentation>
1590     </xs:annotation>
1591   </xs:enumeration>
1592   <xs:enumeration value="freezingFog">
1593     <xs:annotation>
1594       <xs:documentation>Fog, in conjunction with sub-zero air temperatures
↪causing possible freezing of road surface.</xs:documentation>
1595     </xs:annotation>
1596   </xs:enumeration>
1597   <xs:enumeration value="frost">
1598     <xs:annotation>
1599       <xs:documentation>Frost can be expected.</xs:documentation>
1600     </xs:annotation>
1601   </xs:enumeration>
1602   <xs:enumeration value="gales">
1603     <xs:annotation>
1604       <xs:documentation>Winds between 60 km/h and 90 km/h.</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

1605     </xs:annotation>
1606 </xs:enumeration>
1607 <xs:enumeration value="gustyWinds">
1608   <xs:annotation>
1609     <xs:documentation>Constantly varying winds, significant at times.</
↪xs:documentation>
1610   </xs:annotation>
1611 </xs:enumeration>
1612 <xs:enumeration value="hail">
1613   <xs:annotation>
1614     <xs:documentation>Falling ice pellets or frozen rain.</xs:documentation>
1615   </xs:annotation>
1616 </xs:enumeration>
1617 <xs:enumeration value="heavyFrost">
1618   <xs:annotation>
1619     <xs:documentation>A thick coating of frost can be expected.</
↪xs:documentation>
1620   </xs:annotation>
1621 </xs:enumeration>
1622 <xs:enumeration value="heavyRain">
1623   <xs:annotation>
1624     <xs:documentation>Heavy rainfall, limiting visibility to 50m or less.</
↪xs:documentation>
1625   </xs:annotation>
1626 </xs:enumeration>
1627 <xs:enumeration value="heavySnowfall">
1628   <xs:annotation>
1629     <xs:documentation>Dense falling snow, limiting visibility to 50m or less.</
↪xs:documentation>
1630   </xs:annotation>
1631 </xs:enumeration>
1632 <xs:enumeration value="hurricaneForceWinds">
1633   <xs:annotation>
1634     <xs:documentation>Winds over 120 km/h.</xs:documentation>
1635   </xs:annotation>
1636 </xs:enumeration>
1637 <xs:enumeration value="lowSunGlare">
1638   <xs:annotation>
1639     <xs:documentation>Difficult visibility conditions created by low elevation,
↪sunlight.</xs:documentation>
1640   </xs:annotation>
1641 </xs:enumeration>
1642 <xs:enumeration value="moderateFog">
1643   <xs:annotation>
1644     <xs:documentation>Misty conditions impairing vision over 100m.</
↪xs:documentation>
1645   </xs:annotation>
1646 </xs:enumeration>
1647 <xs:enumeration value="ozonePollution">
1648   <xs:annotation>
1649     <xs:documentation>High concentrations of ozone are present.</
↪xs:documentation>
1650   </xs:annotation>
1651 </xs:enumeration>
1652 <xs:enumeration value="pollution">
1653   <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

1654      <xs:documentation>Pollution of an unspecified nature.</xs:documentation>
1655    </xs:annotation>
1656  </xs:enumeration>
1657  <xs:enumeration value="patchyFog">
1658    <xs:annotation>
1659      <xs:documentation>Fog, in which intermittent areas of dense fog may be_
1660    ↪ encountered.</xs:documentation>
1661    </xs:annotation>
1662  </xs:enumeration>
1663  <xs:enumeration value="precipitationInTheArea">
1664    <xs:annotation>
1665      <xs:documentation>Unspecified precipitation is falling on the area.</
1666    ↪ xs:documentation>
1667    </xs:annotation>
1668  </xs:enumeration>
1669  <xs:enumeration value="rain">
1670    <xs:annotation>
1671      <xs:documentation>Rain, visibility more than 50m.</xs:documentation>
1672    </xs:annotation>
1673  </xs:enumeration>
1674  <xs:enumeration value="rainChangingToSnow">
1675    <xs:annotation>
1676      <xs:documentation>Falling rain is changing to snow.</xs:documentation>
1677    </xs:annotation>
1678  </xs:enumeration>
1679  <xs:enumeration value="sandStorms">
1680    <xs:annotation>
1681      <xs:documentation>Sand blowing across the roadway causing significantly_
1682    ↪ reduced visibility.</xs:documentation>
1683    </xs:annotation>
1684  </xs:enumeration>
1685  <xs:enumeration value="severeExhaustPollution">
1686    <xs:annotation>
1687      <xs:documentation>Pollution from exhaust fumes has reached a level_
1688    ↪ sufficient to cause concern.</xs:documentation>
1689    </xs:annotation>
1690  </xs:enumeration>
1691  <xs:enumeration value="severeSmog">
1692    <xs:annotation>
1693      <xs:documentation>Environmental warning of very poor air quality resulting_
1694    ↪ from smog.</xs:documentation>
1695    </xs:annotation>
1696  </xs:enumeration>
1697  <xs:enumeration value="showers">
1698    <xs:annotation>
1699      <xs:documentation>Light rain or intermittent rain.</xs:documentation>
1700    </xs:annotation>
1701  </xs:enumeration>
1702  <xs:enumeration value="sleet">
1703    <xs:annotation>
1704      <xs:documentation>Rain mingled with snow or hail.</xs:documentation>
1705    </xs:annotation>
1706  </xs:enumeration>
1707  <xs:enumeration value="smogAlert">
1708    <xs:annotation>
1709      <xs:documentation>Environmental warning of poor air quality resulting from_
1710    ↪

```

(continues on next page)

(continued from previous page)

```

1705   </xs:documentation>
1706   </xs:annotation>
1707   </xs:enumeration>
1708   <xs:enumeration value="smokeHazard">
1709     <xs:annotation>
1710       <xs:documentation>Smoke drifting across the roadway causing significantly
1711       </xs:documentation>
1712     </xs:annotation>
1713   </xs:enumeration>
1714   <xs:enumeration value="snowChangingToRain">
1715     <xs:annotation>
1716       <xs:documentation>Falling snow is changing to rain.</xs:documentation>
1717     </xs:annotation>
1718   </xs:enumeration>
1719   <xs:enumeration value="snowfall">
1720     <xs:annotation>
1721       <xs:documentation>Falling snow, visibility more than 50m.</xs:documentation>
1722     </xs:annotation>
1723   </xs:enumeration>
1724   <xs:enumeration value="sprayHazard">
1725     <xs:annotation>
1726       <xs:documentation>Reduced visibility resulting from spray created by moving
1727       </xs:documentation>
1728     </xs:annotation>
1729   </xs:enumeration>
1730   <xs:enumeration value="stormForceWinds">
1731     <xs:annotation>
1732       <xs:documentation>Winds between 90 km/h and 120 km/h.</xs:documentation>
1733     </xs:annotation>
1734   </xs:enumeration>
1735   <xs:enumeration value="strongGustsOfWind">
1736     <xs:annotation>
1737       <xs:documentation>Constantly varying winds, strong at times.</
1738     </xs:documentation>
1739   </xs:enumeration>
1740   <xs:enumeration value="strongWinds">
1741     <xs:annotation>
1742       <xs:documentation>Winds between 40 km/h and 60 km/h.</xs:documentation>
1743     </xs:annotation>
1744   </xs:enumeration>
1745   <xs:enumeration value="swarmsOfInsects">
1746     <xs:annotation>
1747       <xs:documentation>Large numbers of insects which create a hazard for road
1748       </xs:documentation>
1749     </xs:annotation>
1750   </xs:enumeration>
1751   <xs:enumeration value="temperatureFalling">
1752     <xs:annotation>
1753       <xs:documentation>The temperature is falling significantly.</
1754     </xs:documentation>
1755   </xs:enumeration>
1756   <xs:enumeration value="thunderstorms">
1757     <xs:annotation>
1758       <xs:documentation>Electrical storms, generally with heavy rain.</
1759     </xs:documentation>
1760   </xs:enumeration>

```

(continues on next page)

(continued from previous page)

```

1755 </xs:documentation>
1756 </xs:enumeration>
1757 <xs:enumeration value="tornadoes">
1758   <xs:annotation>
1759     <xs:documentation>Very violent, whirling windstorms affecting narrow strips_
1760 of country.</xs:documentation>
1761   </xs:annotation>
1762 </xs:enumeration>
1763 <xs:enumeration value="veryStrongGustsOfWind">
1764   <xs:annotation>
1765     <xs:documentation>Constantly varying winds, very strong at times.</
1766 </xs:documentation>
1767   </xs:annotation>
1768 </xs:enumeration>
1769 <xs:enumeration value="visibilityReduced">
1770   <xs:annotation>
1771     <xs:documentation>Environmental conditions causing reduced visibility.</
1772 </xs:documentation>
1773   </xs:annotation>
1774 </xs:enumeration>
1775 <xs:enumeration value="whiteOut">
1776   <xs:annotation>
1777     <xs:documentation>Falling snow in blizzard conditions resulting in very_
1778 reduced visibility.</xs:documentation>
1779   </xs:annotation>
1780 </xs:enumeration>
1781 <xs:enumeration value="winterStorm">
1782   <xs:annotation>
1783     <xs:documentation>Heavy rain, sleet, hail and/or snow in combination with_
1784 strong winds, limiting visibility to 50m or less.</xs:documentation>
1785   </xs:annotation>
1786 </xs:enumeration>
1787 </xs:restriction>
1788 </xs:simpleType>
1789 <xs:complexType name="PrecipitationDetail">
1790   <xs:annotation>
1791     <xs:documentation>Details of precipitation (rain, snow etc.).</xs:documentation>
1792   </xs:annotation>
1793   <xs:sequence>
1794     <xs:element name="precipitationType" type="D2LogicalModel:PrecipitationTypeEnum
1795 minOccurs="0" maxOccurs="1">
1796     <xs:annotation>
1797       <xs:documentation>The type of precipitation which is affecting the driving_
1798 conditions.</xs:documentation>
1799     </xs:annotation>
1800   </xs:element>
1801     <xs:element name="precipitationIntensity" type=
1802 "D2LogicalModel:PrecipitationIntensityValue" minOccurs="0">
1803     <xs:annotation>
1804       <xs:documentation>The height of the precipitation received per unit time.</
1805 </xs:documentation>
1806     </xs:annotation>
1807   </xs:element>
1808     <xs:element name="depositionDepth" type=
1809 "D2LogicalModel:FloatingPointMetreDistanceValue" minOccurs="0">

```

(continues on next page)

(continued from previous page)

```

1800     <xs:annotation>
1801       <xs:documentation>The equivalent depth of the water layer resulting from
↪ precipitation or deposition on a non-porous horizontal surface. Non liquid
↪ precipitation is considered as melted in water form. </xs:documentation>
1802     </xs:annotation>
1803   </xs:element>
1804   <xs:element name="precipitationDetailExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
1805 </xs:sequence>
1806 </xs:complexType>
1807 <xs:complexType name="PrecipitationIntensityValue">
1808   <xs:annotation>
1809     <xs:documentation>A measured or calculated value of the accumulation rate of
↪ precipitation.</xs:documentation>
1810   </xs:annotation>
1811   <xs:complexContent>
1812     <xs:extension base="D2LogicalModel:DataValue">
1813       <xs:sequence>
1814         <xs:element name="millimetresPerHourIntensity" type=
↪ "D2LogicalModel:IntensityMillimetresPerHour" minOccurs="1" maxOccurs="1">
1815           <xs:annotation>
1816             <xs:documentation>A value of precipitation intensity expressed in units
↪ of millimetres per hour.</xs:documentation>
1817           </xs:annotation>
1818           </xs:element>
1819           <xs:element name="precipitationIntensityValueExtension" type=
↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
1820         </xs:sequence>
1821       </xs:extension>
1822     </xs:complexContent>
1823   </xs:complexType>
1824   <xs:simpleType name="PrecipitationTypeEnum">
1825     <xs:annotation>
1826       <xs:documentation>Types of precipitation.</xs:documentation>
1827     </xs:annotation>
1828     <xs:restriction base="xs:string">
1829       <xs:enumeration value="drizzle">
1830         <xs:annotation>
1831           <xs:documentation>Light, fine rain.</xs:documentation>
1832         </xs:annotation>
1833       </xs:enumeration>
1834       <xs:enumeration value="freezingRain">
1835         <xs:annotation>
1836           <xs:documentation>Freezing rain.</xs:documentation>
1837         </xs:annotation>
1838       </xs:enumeration>
1839       <xs:enumeration value="hail">
1840         <xs:annotation>
1841           <xs:documentation>Small balls of ice and compacted snow.</xs:documentation>
1842         </xs:annotation>
1843       </xs:enumeration>
1844       <xs:enumeration value="rain">
1845         <xs:annotation>
1846           <xs:documentation>Rain.</xs:documentation>
1847         </xs:annotation>
1848       </xs:enumeration>

```

(continues on next page)

(continued from previous page)

```

1849     <xs:enumeration value="sleet">
1850       <xs:annotation>
1851         <xs:documentation>Wet snow mixed with rain.</xs:documentation>
1852       </xs:annotation>
1853     </xs:enumeration>
1854     <xs:enumeration value="snow">
1855       <xs:annotation>
1856         <xs:documentation>Snow.</xs:documentation>
1857       </xs:annotation>
1858     </xs:enumeration>
1859   </xs:restriction>
1860 </xs:simpleType>
1861 <xs:simpleType name="ProbabilityOfOccurrenceEnum">
1862   <xs:annotation>
1863     <xs:documentation>Levels of confidence that the sender has in the information,
1864     ↪ordered {certain, probable, risk of}.</xs:documentation>
1865   </xs:annotation>
1866   <xs:restriction base="xs:string">
1867     <xs:enumeration value="certain">
1868       <xs:annotation>
1869         <xs:documentation>The source is completely certain of the occurrence of the
1870         ↪situation record version content.</xs:documentation>
1871       </xs:annotation>
1872     </xs:enumeration>
1873     <xs:enumeration value="probable">
1874       <xs:annotation>
1875         <xs:documentation>The source has a reasonably high level of confidence of
1876         ↪the occurrence of the situation record version content.</xs:documentation>
1877       </xs:annotation>
1878     </xs:enumeration>
1879     <xs:enumeration value="riskOf">
1880       <xs:annotation>
1881         <xs:documentation>The source has a moderate level of confidence of the
1882         ↪occurrence of the situation record version content.</xs:documentation>
1883       </xs:annotation>
1884     </xs:enumeration>
1885   </xs:restriction>
1886 </xs:simpleType>
1887 <xs:complexType name="RoadConditions" abstract="true">
1888   <xs:annotation>
1889     <xs:documentation>Conditions of the road surface which may affect driving
1890     ↪conditions. These may be related to the weather (e.g. ice, snow etc.) or to other
1891     ↪conditions (e.g. oil, mud, leaves etc. on the road)</xs:documentation>
1892   </xs:annotation>
1893   <xs:complexContent>
1894     <xs:extension base="D2LogicalModel:Conditions">
1895       <xs:sequence>
1896         <xs:element name="roadConditionsExtension" type="D2LogicalModel:_
1897         ↪ExtensionType" minOccurs="0" />
1898       </xs:sequence>
1899     </xs:extension>
1900   </xs:complexContent>
1901 </xs:complexType>
1902 <xs:complexType name="RoadSurfaceConditionMeasurements">
1903   <xs:annotation>
1904     <xs:documentation>Measurements of the road surface condition which relate
1905     ↪

```

(continues on next page)

(continued from previous page)

```

1898     </xs:annotation>
1899     <xs:sequence>
1900       <xs:element name="roadSurfaceTemperature" type="D2LogicalModel:TemperatureValue
1901       <xs:annotation>
1902         <xs:documentation>The temperature measured on the road surface.</
1903       </xs:annotation>
1904       </xs:element>
1905       <xs:element name="protectionTemperature" type="D2LogicalModel:TemperatureValue"
1906       <xs:annotation>
1907         <xs:documentation>The road surface temperature down to which the surface is
1908       </xs:annotation>
1909       </xs:element>
1910       <xs:element name="deIcingApplicationRate" type=
1911       <xs:annotation>
1912         <xs:documentation>Indicates the rate at which de-icing agents have been
1913       </xs:annotation>
1914       </xs:element>
1915       <xs:element name="deIcingConcentration" type=
1916       <xs:annotation>
1917         <xs:documentation>Indicates the concentration of de-icing agent present in
1918       </xs:annotation>
1919       </xs:element>
1920       <xs:element name="depthOfSnow" type=
1921       <xs:annotation>
1922         <xs:documentation>The depth of snow recorded on the road surface.</
1923       </xs:annotation>
1924       </xs:element>
1925       <xs:element name="waterFilmThickness" type=
1926       <xs:annotation>
1927         <xs:documentation>The depth of standing water to be found on the road
1928       </xs:annotation>
1929       </xs:element>
1930       <xs:element name="roadSurfaceConditionMeasurementsExtension" type=
1931       </xs:sequence>
1932     </xs:complexType>
1933     <xs:complexType name="Situation">
1934       <xs:annotation>
1935         <xs:documentation>An identifiable instance of a traffic/travel situation
1936         </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

1937     <xs:sequence>
1938       <xs:element name="situationVersionTime" type="D2LogicalModel:DateTime"
1939         ↪minOccurs="0" maxOccurs="1">
1940         <xs:annotation>
1941           <xs:documentation>The date/time that this current version of the Situation
1942             ↪was written into the database of the supplier which is involved in the data
1943             ↪exchange. Identity and version of the situation are defined by the class stereotype
1944             ↪implementation.</xs:documentation>
1945         </xs:annotation>
1946       </xs:element>
1947       <xs:element name="headerInformation" type="D2LogicalModel:HeaderInformation" />
1948       <xs:element name="situationRecord" type="D2LogicalModel:SituationRecord"
1949         ↪maxOccurs="unbounded" />
1950       <xs:element name="situationExtension" type="D2LogicalModel:_ExtensionType"
1951         ↪minOccurs="0" />
1952     </xs:sequence>
1953     <xs:attribute name="id" type="xs:string" use="required" />
1954     <xs:attribute name="version" type="xs:string" use="required" />
1955 </xs:complexType>
1956 <xs:complexType name="SituationPublication">
1957   <xs:annotation>
1958     <xs:documentation>A publication containing zero or more traffic/travel
1959     ↪situations.</xs:documentation>
1960   </xs:annotation>
1961   <xs:complexContent>
1962     <xs:extension base="D2LogicalModel:PayloadPublication">
1963       <xs:sequence>
1964         <xs:element name="situation" type="D2LogicalModel:Situation" minOccurs="0"
1965         ↪maxOccurs="unbounded" />
1966         <xs:element name="situationPublicationExtension" type="D2LogicalModel:_
1967         ↪ExtensionType" minOccurs="0" />
1968       </xs:sequence>
1969     </xs:extension>
1970   </xs:complexContent>
1971 </xs:complexType>
1972 <xs:complexType name="SituationRecord" abstract="true">
1973   <xs:annotation>
1974     <xs:documentation>An identifiable versioned instance of a single record/element
1975     ↪within a situation.</xs:documentation>
1976   </xs:annotation>
1977   <xs:sequence>
1978     <xs:element name="situationRecordCreationTime" type="D2LogicalModel:DateTime"
1979     ↪minOccurs="1" maxOccurs="1">
1980     <xs:annotation>
1981       <xs:documentation>The date/time that the SituationRecord object (the first
1982       ↪version of the record) was created by the original supplier.</xs:documentation>
1983     </xs:annotation>
1984     </xs:element>
1985     <xs:element name="situationRecordVersionTime" type="D2LogicalModel:DateTime"
1986     ↪minOccurs="1" maxOccurs="1">
1987     <xs:annotation>
1988       <xs:documentation>The date/time that this current version of the
1989       ↪SituationRecord within the situation was written into the database of the supplier
1990       ↪which is involved in the data exchange. Identity and version of record are defined
1991       ↪by the class stereotype implementation.</xs:documentation>
1992     </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

1977     </xs:element>
1978     <xs:element name="probabilityOfOccurrence" type=
1979 ↪ "D2LogicalModel:ProbabilityOfOccurrenceEnum" minOccurs="1" maxOccurs="1">
1980         <xs:annotation>
1981             <xs:documentation>An assessment of the degree of likelihood that the
1982 ↪ reported event will occur.</xs:documentation>
1983         </xs:annotation>
1984     </xs:element>
1985     <xs:element name="source" type="D2LogicalModel:Source" minOccurs="0" />
1986     <xs:element name="validity" type="D2LogicalModel:Validity" />
1987     <xs:element name="cause" type="D2LogicalModel:Cause" minOccurs="0" />
1988     <xs:element name="generalPublicComment" type="D2LogicalModel:Comment" minOccurs=
1989 ↪ "0" maxOccurs="unbounded">
1990         <xs:annotation>
1991             <xs:documentation>A comment which may be freely distributed to the general
1992 ↪ public</xs:documentation>
1993         </xs:annotation>
1994     </xs:element>
1995     <xs:element name="groupOfLocations" type="D2LogicalModel:GroupOfLocations" />
1996     <xs:element name="situationRecordExtension" type="D2LogicalModel:_ExtensionType
1997 ↪ " minOccurs="0" />
1998     </xs:sequence>
1999     <xs:attribute name="id" type="xs:string" use="required" />
2000     <xs:attribute name="version" type="xs:string" use="required" />
2001 </xs:complexType>
2002 <xs:complexType name="Source">
2003     <xs:annotation>
2004         <xs:documentation>Details of the source from which the information was obtained.
2005 ↪ </xs:documentation>
2006     </xs:annotation>
2007     <xs:sequence>
2008         <xs:element name="sourceCountry" type="D2LogicalModel:CountryEnum" minOccurs="0
2009 ↪ " maxOccurs="1">
2010             <xs:annotation>
2011                 <xs:documentation>ISO 3166-1 two character country code of the source of
2012 ↪ the information.</xs:documentation>
2013             </xs:annotation>
2014         </xs:element>
2015         <xs:element name="sourceIdentification" type="D2LogicalModel:String" minOccurs=
2016 ↪ "0" maxOccurs="1">
2017             <xs:annotation>
2018                 <xs:documentation>Identifier of the organisation or the traffic equipment
2019 ↪ which has produced the information relating to this version of the information.</
2020 ↪ xs:documentation>
2021             </xs:annotation>
2022         </xs:element>
2023         <xs:element name="sourceName" type="D2LogicalModel:MultilingualString"
2024 ↪ minOccurs="0" maxOccurs="1">
2025             <xs:annotation>
2026                 <xs:documentation>The name of the organisation which has produced the
2027 ↪ information relating to this version of the information.</xs:documentation>
2028             </xs:annotation>
2029         </xs:element>
2030         <xs:element name="sourceType" type="D2LogicalModel:SourceTypeEnum" minOccurs="0
2031 ↪ " maxOccurs="1">
2032             <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

2019     <xs:documentation>Information about the technology used for measuring the
↪data or the method used for obtaining qualitative descriptions relating to this
↪version of the information.</xs:documentation>
2020     </xs:annotation>
2021     </xs:element>
2022     <xs:element name="reliable" type="D2LogicalModel:Boolean" minOccurs="0"
↪maxOccurs="1">
2023         <xs:annotation>
2024             <xs:documentation>An indication as to whether the source deems the
↪associated information to be reliable/correct. "True" indicates it is deemed
↪reliable.</xs:documentation>
2025         </xs:annotation>
2026         </xs:element>
2027         <xs:element name="sourceExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />
2028     </xs:sequence>
2029 </xs:complexType>
2030 <xs:simpleType name="SourceTypeEnum">
2031     <xs:annotation>
2032         <xs:documentation>Type of sources from which situation information may be
↪derived.</xs:documentation>
2033     </xs:annotation>
2034     <xs:restriction base="xs:string">
2035         <xs:enumeration value="automobileClubPatrol">
2036             <xs:annotation>
2037                 <xs:documentation>A patrol of an automobile club.</xs:documentation>
2038             </xs:annotation>
2039         </xs:enumeration>
2040         <xs:enumeration value="cameraObservation">
2041             <xs:annotation>
2042                 <xs:documentation>A camera observation (either still or video camera).</
↪xs:documentation>
2043             </xs:annotation>
2044         </xs:enumeration>
2045         <xs:enumeration value="freightVehicleOperator">
2046             <xs:annotation>
2047                 <xs:documentation>An operator of freight vehicles.</xs:documentation>
2048             </xs:annotation>
2049         </xs:enumeration>
2050         <xs:enumeration value="inductionLoopMonitoringStation">
2051             <xs:annotation>
2052                 <xs:documentation>A station dedicated to the monitoring of the road network
↪by processing inductive loop information.</xs:documentation>
2053             </xs:annotation>
2054         </xs:enumeration>
2055         <xs:enumeration value="infraredMonitoringStation">
2056             <xs:annotation>
2057                 <xs:documentation>A station dedicated to the monitoring of the road network
↪by processing infrared image information.</xs:documentation>
2058             </xs:annotation>
2059         </xs:enumeration>
2060         <xs:enumeration value="microwaveMonitoringStation">
2061             <xs:annotation>
2062                 <xs:documentation>A station dedicated to the monitoring of the road network
↪by processing microwave information.</xs:documentation>
2063             </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

2064     </xs:enumeration>
2065     <xs:enumeration value="mobileTelephoneCaller">
2066         <xs:annotation>
2067             <xs:documentation>A caller using a mobile telephone (who may or may not be
↪ on the road network).</xs:documentation>
2068         </xs:annotation>
2069     </xs:enumeration>
2070     <xs:enumeration value="nonPoliceEmergencyServicePatrol">
2071         <xs:annotation>
2072             <xs:documentation>Emergency service patrols other than police.</
↪ xs:documentation>
2073         </xs:annotation>
2074     </xs:enumeration>
2075     <xs:enumeration value="otherInformation">
2076         <xs:annotation>
2077             <xs:documentation>Other sources of information.</xs:documentation>
2078         </xs:annotation>
2079     </xs:enumeration>
2080     <xs:enumeration value="otherOfficialVehicle">
2081         <xs:annotation>
2082             <xs:documentation>Personnel from a vehicle belonging to the road operator
↪ or authority or any emergency service, including authorised breakdown service
↪ organisations.</xs:documentation>
2083         </xs:annotation>
2084     </xs:enumeration>
2085     <xs:enumeration value="policePatrol">
2086         <xs:annotation>
2087             <xs:documentation>A police patrol.</xs:documentation>
2088         </xs:annotation>
2089     </xs:enumeration>
2090     <xs:enumeration value="privateBreakdownService">
2091         <xs:annotation>
2092             <xs:documentation>A private breakdown service.</xs:documentation>
2093         </xs:annotation>
2094     </xs:enumeration>
2095     <xs:enumeration value="publicAndPrivateUtilities">
2096         <xs:annotation>
2097             <xs:documentation>A utility organisation, either public or private.</
↪ xs:documentation>
2098         </xs:annotation>
2099     </xs:enumeration>
2100     <xs:enumeration value="registeredMotoristObserver">
2101         <xs:annotation>
2102             <xs:documentation>A motorist who is an officially registered observer.</
↪ xs:documentation>
2103         </xs:annotation>
2104     </xs:enumeration>
2105     <xs:enumeration value="roadAuthorities">
2106         <xs:annotation>
2107             <xs:documentation>A road authority.</xs:documentation>
2108         </xs:annotation>
2109     </xs:enumeration>
2110     <xs:enumeration value="roadOperatorPatrol">
2111         <xs:annotation>
2112             <xs:documentation>A patrol of the road operator or authority.</
↪ xs:documentation>

```

(continues on next page)

(continued from previous page)

```

2113     </xs:annotation>
2114 </xs:enumeration>
2115 <xs:enumeration value="roadsideTelephoneCaller">
2116   <xs:annotation>
2117     <xs:documentation>A caller who is using an emergency roadside telephone.</
↪xs:documentation>
2118   </xs:annotation>
2119 </xs:enumeration>
2120 <xs:enumeration value="spotterAircraft">
2121   <xs:annotation>
2122     <xs:documentation>A spotter aircraft of an organisation specifically
↪assigned to the monitoring of the traffic network.</xs:documentation>
2123   </xs:annotation>
2124 </xs:enumeration>
2125 <xs:enumeration value="trafficMonitoringStation">
2126   <xs:annotation>
2127     <xs:documentation>A station, usually automatic, dedicated to the monitoring
↪of the road network.</xs:documentation>
2128   </xs:annotation>
2129 </xs:enumeration>
2130 <xs:enumeration value="transitOperator">
2131   <xs:annotation>
2132     <xs:documentation>An operator of a transit service, e.g. bus link operator.
↪</xs:documentation>
2133   </xs:annotation>
2134 </xs:enumeration>
2135 <xs:enumeration value="vehicleProbeMeasurement">
2136   <xs:annotation>
2137     <xs:documentation>A specially equipped vehicle used to provide measurements.
↪</xs:documentation>
2138   </xs:annotation>
2139 </xs:enumeration>
2140 <xs:enumeration value="videoProcessingMonitoringStation">
2141   <xs:annotation>
2142     <xs:documentation>A station dedicated to the monitoring of the road network
↪by processing video image information.</xs:documentation>
2143   </xs:annotation>
2144 </xs:enumeration>
2145 </xs:restriction>
2146 </xs:simpleType>
2147 <xs:complexType name="SpeedValue">
2148   <xs:annotation>
2149     <xs:documentation>A measured or calculated value of speed.</xs:documentation>
2150   </xs:annotation>
2151   <xs:complexContent>
2152     <xs:extension base="D2LogicalModel:DataValue">
2153       <xs:sequence>
2154         <xs:element name="speed" type="D2LogicalModel:KilometresPerHour" minOccurs=
↪"1" maxOccurs="1">
2155           <xs:annotation>
2156             <xs:documentation>A value of speed expressed in kilometres per hour.</
↪xs:documentation>
2157           </xs:annotation>
2158         </xs:element>
2159         <xs:element name="speedValueExtension" type="D2LogicalModel:_ExtensionType"
↪minOccurs="0" />

```

(continues on next page)

(continued from previous page)

```

2160     </xs:sequence>
2161   </xs:extension>
2162 </xs:complexContent>
2163 </xs:complexType>
2164 <xs:simpleType name="String">
2165   <xs:annotation>
2166     <xs:documentation>A character string whose value space is the set of finite-
↪ length sequences of characters. Every character has a corresponding Universal
↪ Character Set code point (as defined in ISO/IEC 10646), which is an integer.</
↪ xs:documentation>
2167   </xs:annotation>
2168   <xs:restriction base="xs:string">
2169     <xs:maxLength value="1024" />
2170   </xs:restriction>
2171 </xs:simpleType>
2172 <xs:complexType name="Temperature">
2173   <xs:annotation>
2174     <xs:documentation>Details of atmospheric temperature.</xs:documentation>
2175   </xs:annotation>
2176   <xs:sequence>
2177     <xs:element name="airTemperature" type="D2LogicalModel:TemperatureValue"
↪ minOccurs="0">
2178       <xs:annotation>
2179         <xs:documentation>The air temperature measured in the shade between 1.5 and
↪ 2 metres above ground level.</xs:documentation>
2180       </xs:annotation>
2181     </xs:element>
2182     <xs:element name="dewPointTemperature" type="D2LogicalModel:TemperatureValue"
↪ minOccurs="0">
2183       <xs:annotation>
2184         <xs:documentation>The temperature to which the air would have to cool (at
↪ constant pressure and water vapour content) in order to reach saturation.</
↪ xs:documentation>
2185       </xs:annotation>
2186     </xs:element>
2187     <xs:element name="maximumTemperature" type="D2LogicalModel:TemperatureValue"
↪ minOccurs="0">
2188       <xs:annotation>
2189         <xs:documentation>The maximum temperature during the forecast or
↪ measurement period.</xs:documentation>
2190       </xs:annotation>
2191     </xs:element>
2192     <xs:element name="minimumTemperature" type="D2LogicalModel:TemperatureValue"
↪ minOccurs="0">
2193       <xs:annotation>
2194         <xs:documentation>The minimum temperature during the forecast or
↪ measurement period.</xs:documentation>
2195       </xs:annotation>
2196     </xs:element>
2197     <xs:element name="temperatureExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2198   </xs:sequence>
2199 </xs:complexType>
2200 <xs:simpleType name="TemperatureCelsius">
2201   <xs:annotation>
2202     <xs:documentation>A measure of temperature defined in degrees Celsius.</

```

(continues on next page)

(continued from previous page)

```

2203 </xs:documentation>
2204 <xs:restriction base="D2LogicalModel:Float" />
2205 </xs:simpleType>
2206 <xs:complexType name="TemperatureValue">
2207   <xs:annotation>
2208     <xs:documentation>A measured or calculated value of temperature.</
2209   </xs:annotation>
2210   <xs:complexContent>
2211     <xs:extension base="D2LogicalModel:DataValue">
2212       <xs:sequence>
2213         <xs:element name="temperature" type="D2LogicalModel:TemperatureCelsius"
2214           minOccurs="1" maxOccurs="1">
2215           <xs:annotation>
2216             <xs:documentation>A value of temperature expressed in degrees Celsius.</
2217           </xs:annotation>
2218           </xs:element>
2219           <xs:element name="temperatureValueExtension" type="D2LogicalModel:_
2220             ExtensionType" minOccurs="0" />
2221         </xs:sequence>
2222       </xs:extension>
2223     </xs:complexContent>
2224   </xs:complexType>
2225   <xs:complexType name="TrafficElement" abstract="true">
2226     <xs:annotation>
2227       <xs:documentation>An event which is not planned by the traffic operator, which
2228       is affecting, or has the potential to affect traffic flow.</xs:documentation>
2229     </xs:annotation>
2230     <xs:complexContent>
2231       <xs:extension base="D2LogicalModel:SituationRecord">
2232         <xs:sequence>
2233           <xs:element name="trafficElementExtension" type="D2LogicalModel:_
2234             ExtensionType" minOccurs="0" />
2235         </xs:sequence>
2236       </xs:extension>
2237     </xs:complexContent>
2238   </xs:complexType>
2239   <xs:simpleType name="TrafficStatusEnum">
2240     <xs:annotation>
2241       <xs:documentation>List of terms used to describe traffic conditions.</
2242     </xs:annotation>
2243     <xs:restriction base="xs:string">
2244       <xs:enumeration value="impossible">
2245         <xs:annotation>
2246           <xs:documentation>Traffic in the specified direction is completely
2247           congested, effectively at a standstill, making driving impossible.</
2248         </xs:annotation>
2249       </xs:enumeration>
2250       <xs:enumeration value="congested">
2251         <xs:annotation>
2252           <xs:documentation>Traffic in the specified direction is congested making
2253           driving very slow and difficult.</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

2248     </xs:annotation>
2249   </xs:enumeration>
2250   <xs:enumeration value="heavy">
2251     <xs:annotation>
2252       <xs:documentation>Traffic in the specified direction is heavier than usual.
↳ making driving conditions more difficult than normal.</xs:documentation>
2253     </xs:annotation>
2254   </xs:enumeration>
2255   <xs:enumeration value="freeFlow">
2256     <xs:annotation>
2257       <xs:documentation>Traffic in the specified direction is free flowing.</
↳ xs:documentation>
2258     </xs:annotation>
2259   </xs:enumeration>
2260   <xs:enumeration value="unknown">
2261     <xs:annotation>
2262       <xs:documentation>Traffic conditions are unknown.</xs:documentation>
2263     </xs:annotation>
2264   </xs:enumeration>
2265 </xs:restriction>
2266 </xs:simpleType>
2267 <xs:complexType name="TrafficStatusValue">
2268   <xs:annotation>
2269     <xs:documentation>A measured or calculated value of the status of traffic
↳ conditions on a section of road in a specified direction.</xs:documentation>
2270   </xs:annotation>
2271   <xs:complexContent>
2272     <xs:extension base="D2LogicalModel:DataValue">
2273       <xs:sequence>
2274         <xs:element name="trafficStatusValue" type="D2LogicalModel:TrafficStatusEnum
↳ " minOccurs="1" maxOccurs="1">
2275           <xs:annotation>
2276             <xs:documentation>A status value of traffic conditions on the
↳ identified section of road in the specified direction.</xs:documentation>
2277           </xs:annotation>
2278         </xs:element>
2279         <xs:element name="trafficStatusValueExtension" type="D2LogicalModel:_
↳ ExtensionType" minOccurs="0" />
2280       </xs:sequence>
2281     </xs:extension>
2282   </xs:complexContent>
2283 </xs:complexType>
2284 <xs:simpleType name="UrgencyEnum">
2285   <xs:annotation>
2286     <xs:documentation>Degrees of urgency that a receiving client should associate
↳ with the disseminate of the information contained in the publication.</
↳ xs:documentation>
2287   </xs:annotation>
2288   <xs:restriction base="xs:string">
2289     <xs:enumeration value="extremelyUrgent">
2290       <xs:annotation>
2291         <xs:documentation>Dissemination of the information is extremely urgent.</
↳ xs:documentation>
2292       </xs:annotation>
2293     </xs:enumeration>
2294     <xs:enumeration value="urgent">

```

(continues on next page)

(continued from previous page)

```

2295     <xs:annotation>
2296       <xs:documentation>Dissemination of the information is urgent.</
↪ <xs:documentation>
2297     </xs:annotation>
2298   </xs:enumeration>
2299   <xs:enumeration value="normalUrgency">
2300     <xs:annotation>
2301       <xs:documentation>Dissemination of the information is of normal urgency.</
↪ <xs:documentation>
2302     </xs:annotation>
2303   </xs:enumeration>
2304 </xs:restriction>
2305 </xs:simpleType>
2306 <xs:complexType name="Validity">
2307   <xs:annotation>
2308     <xs:documentation>Specification of validity, either explicitly or by a validity_
↪ time period specification which may be discontinuous.</xs:documentation>
2309   </xs:annotation>
2310   <xs:sequence>
2311     <xs:element name="validityStatus" type="D2LogicalModel:ValidityStatusEnum"
↪ minOccurs="1" maxOccurs="1">
2312       <xs:annotation>
2313         <xs:documentation>Specification of validity, either explicitly overriding_
↪ the validity time specification or confirming it.</xs:documentation>
2314       </xs:annotation>
2315     </xs:element>
2316     <xs:element name="validityTimeSpecification" type="D2LogicalModel:OverallPeriod
↪ ">
2317       <xs:annotation>
2318         <xs:documentation>A specification of periods of validity defined by overall_
↪ bounding start and end times and the possible intersection of valid periods with_
↪ exception periods (exception periods overriding valid periods).</xs:documentation>
2319       </xs:annotation>
2320     </xs:element>
2321     <xs:element name="validityExtension" type="D2LogicalModel:_ExtensionType"
↪ minOccurs="0" />
2322   </xs:sequence>
2323 </xs:complexType>
2324 <xs:simpleType name="ValidityStatusEnum">
2325   <xs:annotation>
2326     <xs:documentation>Values of validity status that can be assigned to a described_
↪ event, action or item.</xs:documentation>
2327   </xs:annotation>
2328   <xs:restriction base="xs:string">
2329     <xs:enumeration value="active">
2330       <xs:annotation>
2331         <xs:documentation>The described event, action or item is currently active_
↪ regardless of the definition of the validity time specification.</xs:documentation>
2332       </xs:annotation>
2333     </xs:enumeration>
2334     <xs:enumeration value="suspended">
2335       <xs:annotation>
2336         <xs:documentation>The described event, action or item is currently_
↪ suspended, that is inactive, regardless of the definition of the validity time_
↪ specification.</xs:documentation>
2337       </xs:annotation>

```

(continues on next page)

(continued from previous page)

```

2338     </xs:enumeration>
2339     <xs:enumeration value="definedByValidityTimeSpec">
2340       <xs:annotation>
2341         <xs:documentation>The validity status of the described event, action or
↪ item is in accordance with the definition of the validity time specification.</
↪ xs:documentation>
2342       </xs:annotation>
2343     </xs:enumeration>
2344   </xs:restriction>
2345 </xs:simpleType>
2346 <xs:complexType name="VehicleCountValue">
2347   <xs:annotation>
2348     <xs:documentation>A measured or calculated value of absolute count of vehicles.
↪ within a specified period of time expressed as non negative integer.</
↪ xs:documentation>
2349   </xs:annotation>
2350   <xs:complexContent>
2351     <xs:extension base="D2LogicalModel:DataValue">
2352       <xs:sequence>
2353         <xs:element name="vehicleCount" type="D2LogicalModel:NonNegativeInteger"
↪ minOccurs="1" maxOccurs="1">
2354           <xs:annotation>
2355             <xs:documentation>A measured or calculated absolute count of vehicles.
↪ within a specified period of time expressed as non negative integer.</
↪ xs:documentation>
2356           </xs:annotation>
2357         </xs:element>
2358         <xs:element name="vehicleCountValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2359       </xs:sequence>
2360     </xs:extension>
2361   </xs:complexContent>
2362 </xs:complexType>
2363 <xs:complexType name="VehicleFlowValue">
2364   <xs:annotation>
2365     <xs:documentation>A measured or calculated value of the flow rate of vehicles.</
↪ xs:documentation>
2366   </xs:annotation>
2367   <xs:complexContent>
2368     <xs:extension base="D2LogicalModel:DataValue">
2369       <xs:sequence>
2370         <xs:element name="vehicleFlowRate" type="D2LogicalModel:VehiclesPerHour"
↪ minOccurs="1" maxOccurs="1">
2371           <xs:annotation>
2372             <xs:documentation>A value of vehicle flow rate expressed in vehicles.
↪ per hour.</xs:documentation>
2373           </xs:annotation>
2374         </xs:element>
2375         <xs:element name="vehicleFlowValueExtension" type="D2LogicalModel:_
↪ ExtensionType" minOccurs="0" />
2376       </xs:sequence>
2377     </xs:extension>
2378   </xs:complexContent>
2379 </xs:complexType>
2380 <xs:simpleType name="VehiclesPerHour">
2381   <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

2382     <xs:documentation>Vehicles per hour.</xs:documentation>
2383   </xs:annotation>
2384   <xs:restriction base="D2LogicalModel:NonNegativeInteger" />
2385 </xs:simpleType>
2386 <xs:complexType name="VersionedReference">
2387   <xs:attribute name="id" type="xs:string" use="required" />
2388   <xs:attribute name="version" type="xs:string" use="required" />
2389 </xs:complexType>
2390 <xs:complexType name="Visibility">
2391   <xs:annotation>
2392     <xs:documentation>Details of atmospheric visibility.</xs:documentation>
2393   </xs:annotation>
2394   <xs:sequence>
2395     <xs:element name="minimumVisibilityDistance" type=
2396 ↪ "D2LogicalModel:IntegerMetreDistanceValue">
2397       <xs:annotation>
2398         <xs:documentation>The minimum distance, measured or estimated, beyond which
2399 ↪ drivers may be unable to clearly see a vehicle or an obstacle.</xs:documentation>
2400       </xs:annotation>
2401     </xs:element>
2402     <xs:element name="visibilityExtension" type="D2LogicalModel:_ExtensionType"
2403 ↪ minOccurs="0" />
2404   </xs:sequence>
2405 </xs:complexType>
2406 <xs:complexType name="WeatherRelatedRoadConditions">
2407   <xs:annotation>
2408     <xs:documentation>Road surface conditions that are related to the weather which
2409 ↪ may affect the driving conditions, such as ice, snow or water.</xs:documentation>
2410   </xs:annotation>
2411   <xs:complexContent>
2412     <xs:extension base="D2LogicalModel:RoadConditions">
2413       <xs:sequence>
2414         <xs:element name="weatherRelatedRoadConditionType" type=
2415 ↪ "D2LogicalModel:WeatherRelatedRoadConditionTypeEnum" minOccurs="1" maxOccurs=
2416 ↪ "unbounded">
2417           <xs:annotation>
2418             <xs:documentation>The type of road surface condition that is related to
2419 ↪ the weather which is affecting the driving conditions.</xs:documentation>
2420           </xs:annotation>
2421         </xs:element>
2422         <xs:element name="roadSurfaceConditionMeasurements" type=
2423 ↪ "D2LogicalModel:RoadSurfaceConditionMeasurements" minOccurs="0" />
2424         <xs:element name="weatherRelatedRoadConditionsExtension" type=
2425 ↪ "D2LogicalModel:_ExtensionType" minOccurs="0" />
2426       </xs:sequence>
2427     </xs:extension>
2428   </xs:complexContent>
2429 </xs:complexType>
2430 <xs:simpleType name="WeatherRelatedRoadConditionTypeEnum">
2431   <xs:annotation>
2432     <xs:documentation>Types of road surface conditions which are related to the
2433 ↪ weather.</xs:documentation>
2434   </xs:annotation>
2435   <xs:restriction base="xs:string">
2436     <xs:enumeration value="blackIce">
2437       <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

2428     <xs:documentation>Severe skid risk due to black ice (i.e. clear ice, which
↪ is impossible or very difficult to see).</xs:documentation>
2429     </xs:annotation>
2430   </xs:enumeration>
2431   <xs:enumeration value="deepSnow">
2432     <xs:annotation>
2433       <xs:documentation>Deep snow on the roadway.</xs:documentation>
2434     </xs:annotation>
2435   </xs:enumeration>
2436   <xs:enumeration value="dry">
2437     <xs:annotation>
2438       <xs:documentation>The road surface is dry.</xs:documentation>
2439     </xs:annotation>
2440   </xs:enumeration>
2441   <xs:enumeration value="freezingOfWetRoads">
2442     <xs:annotation>
2443       <xs:documentation>The wet road surface is subject to freezing.</
↪ xs:documentation>
2444     </xs:annotation>
2445   </xs:enumeration>
2446   <xs:enumeration value="freezingPavements">
2447     <xs:annotation>
2448       <xs:documentation>The pavements for pedestrians are subject to freezing.</
↪ xs:documentation>
2449     </xs:annotation>
2450   </xs:enumeration>
2451   <xs:enumeration value="freezingRain">
2452     <xs:annotation>
2453       <xs:documentation>Severe skid risk due to rain falling on sub-zero
↪ temperature road surface and freezing.</xs:documentation>
2454     </xs:annotation>
2455   </xs:enumeration>
2456   <xs:enumeration value="freshSnow">
2457     <xs:annotation>
2458       <xs:documentation>Fresh snow (with little or no traffic yet) on the roadway.
↪ </xs:documentation>
2459     </xs:annotation>
2460   </xs:enumeration>
2461   <xs:enumeration value="ice">
2462     <xs:annotation>
2463       <xs:documentation>Increased skid risk due to ice (of any kind).</
↪ xs:documentation>
2464     </xs:annotation>
2465   </xs:enumeration>
2466   <xs:enumeration value="iceBuildUp">
2467     <xs:annotation>
2468       <xs:documentation>Ice is building up on the roadway causing a serious skid
↪ hazard.</xs:documentation>
2469     </xs:annotation>
2470   </xs:enumeration>
2471   <xs:enumeration value="iceWithWheelBarTracks">
2472     <xs:annotation>
2473       <xs:documentation>Ice on the road frozen in the form of wheel tracks.</
↪ xs:documentation>
2474     </xs:annotation>
2475   </xs:enumeration>

```

(continues on next page)

(continued from previous page)

```

2476     <xs:enumeration value="icyPatches">
2477         <xs:annotation>
2478             <xs:documentation>Severe skid risk due to icy patches (i.e. intermittent_
↪ice on roadway).</xs:documentation>
2479         </xs:annotation>
2480     </xs:enumeration>
2481     <xs:enumeration value="looseSnow">
2482         <xs:annotation>
2483             <xs:documentation>Powdery snow on the road which is subject to being blown_
↪by the wind.</xs:documentation>
2484         </xs:annotation>
2485     </xs:enumeration>
2486     <xs:enumeration value="normalWinterConditionsForPedestrians">
2487         <xs:annotation>
2488             <xs:documentation>Conditions for pedestrians are consistent with those_
↪normally expected in winter.</xs:documentation>
2489         </xs:annotation>
2490     </xs:enumeration>
2491     <xs:enumeration value="packedSnow">
2492         <xs:annotation>
2493             <xs:documentation>Packed snow (heavily trafficked) on the roadway.</
↪xs:documentation>
2494         </xs:annotation>
2495     </xs:enumeration>
2496     <xs:enumeration value="roadSurfaceMelting">
2497         <xs:annotation>
2498             <xs:documentation>The road surface is melting, or has melted due to_
↪abnormally high temperatures.</xs:documentation>
2499         </xs:annotation>
2500     </xs:enumeration>
2501     <xs:enumeration value="slipperyRoad">
2502         <xs:annotation>
2503             <xs:documentation>The road surface is slippery due to bad weather_
↪conditions.</xs:documentation>
2504         </xs:annotation>
2505     </xs:enumeration>
2506     <xs:enumeration value="slushOnRoad">
2507         <xs:annotation>
2508             <xs:documentation>Increased skid risk due to melting snow (slush) on road.</
↪xs:documentation>
2509         </xs:annotation>
2510     </xs:enumeration>
2511     <xs:enumeration value="slushStrings">
2512         <xs:annotation>
2513             <xs:documentation>Melting snow (slush) on the roadway is formed into wheel_
↪tracks.</xs:documentation>
2514         </xs:annotation>
2515     </xs:enumeration>
2516     <xs:enumeration value="snowDrifts">
2517         <xs:annotation>
2518             <xs:documentation>Snow drifting is in progress or patches of deep snow are_
↪present due to earlier drifting.</xs:documentation>
2519         </xs:annotation>
2520     </xs:enumeration>
2521     <xs:enumeration value="snowOnPavement">
2522         <xs:annotation>

```

(continues on next page)

(continued from previous page)

```

2523     <xs:documentation>Snow is on the pedestrian pavement.</xs:documentation>
2524   </xs:annotation>
2525 </xs:enumeration>
2526 <xs:enumeration value="snowOnTheRoad">
2527   <xs:annotation>
2528     <xs:documentation>Snow is lying on the road surface.</xs:documentation>
2529   </xs:annotation>
2530 </xs:enumeration>
2531 <xs:enumeration value="surfaceWater">
2532   <xs:annotation>
2533     <xs:documentation>Water is resting on the roadway which provides an
2534     ↪ increased hazard to vehicles.</xs:documentation>
2535   </xs:annotation>
2536 </xs:enumeration>
2537 <xs:enumeration value="wet">
2538   <xs:annotation>
2539     <xs:documentation>Road surface is wet.</xs:documentation>
2540   </xs:annotation>
2541 </xs:enumeration>
2542 <xs:enumeration value="wetAndIcyRoad">
2543   <xs:annotation>
2544     <xs:documentation>Increased skid risk due to partly thawed, wet road with
2545     ↪ packed snow and ice, or rain falling on packed snow and ice.</xs:documentation>
2546   </xs:annotation>
2547 </xs:enumeration>
2548 <xs:enumeration value="wetIcyPavement">
2549   <xs:annotation>
2550     <xs:documentation>Partly thawed, wet pedestrian pavement with packed snow
2551     ↪ and ice, or rain falling on packed snow and ice.</xs:documentation>
2552   </xs:annotation>
2553 </xs:enumeration>
2554 <xs:enumeration value="other">
2555   <xs:annotation>
2556     <xs:documentation>Other than as defined in this enumeration.</
2557     ↪ xs:documentation>
2558   </xs:annotation>
2559 </xs:enumeration>
2560 </xs:restriction>
2561 </xs:simpleType>
2562 <xs:complexType name="Wind">
2563   <xs:annotation>
2564     <xs:documentation>Wind conditions on the road. </xs:documentation>
2565   </xs:annotation>
2566   <xs:sequence>
2567     <xs:element name="windMeasurementHeight" type=
2568     ↪ "D2LogicalModel:MetresAsNonNegativeInteger" minOccurs="0" maxOccurs="1">
2569       <xs:annotation>
2570         <xs:documentation>The height in metres above the road surface at which the
2571         ↪ wind is measured.</xs:documentation>
2572       </xs:annotation>
2573     </xs:element>
2574     <xs:element name="windSpeed" type="D2LogicalModel:SpeedValue" minOccurs="0">
2575       <xs:annotation>
2576         <xs:documentation>The wind speed averaged over at least 10 minutes,
2577         ↪ measured at a default height of 10 metres (meteorological standard) above the road
2578         ↪ surface, unless measurement height is specified.</xs:documentation>

```

(continues on next page)

(continued from previous page)

```

2571     </xs:annotation>
2572   </xs:element>
2573   <xs:element name="maximumWindSpeed" type="D2LogicalModel:SpeedValue" minOccurs=
↪ "0">
2574     <xs:annotation>
2575       <xs:documentation>The maximum wind speed in a measurement period of 10_
↪ minutes.</xs:documentation>
2576     </xs:annotation>
2577   </xs:element>
2578   <xs:element name="windDirectionBearing" type=
↪ "D2LogicalModel:DirectionBearingValue" minOccurs="0">
2579     <xs:annotation>
2580       <xs:documentation>The average direction from which the wind blows, in terms_
↪ of a bearing measured in degrees (0 - 359).</xs:documentation>
2581     </xs:annotation>
2582   </xs:element>
2583   <xs:element name="windDirectionCompass" type=
↪ "D2LogicalModel:DirectionCompassValue" minOccurs="0">
2584     <xs:annotation>
2585       <xs:documentation>The average direction from which the wind blows, in terms_
↪ of points of the compass.</xs:documentation>
2586     </xs:annotation>
2587   </xs:element>
2588   <xs:element name="windExtension" type="D2LogicalModel:_ExtensionType" minOccurs=
↪ "0" />
2589 </xs:sequence>
2590 </xs:complexType>
2591 </xs:schema>

```

CHANGELOG

version 1.1.0

- added feed type to schema and samples

version 1.0.1

- revision of the documentation, proofreading, changes to examples, some texts regarding locations and concepts have been modified to be more specific.

version 1.0.0

- Initial format and documentation

BIBLIOGRAPHY

- [16157-1] CEN TS 16157-1:2011 Intelligent transport systems – DATEX II data exchange specifications for traffic management and information – Part 1: Context and framework, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:31061&cs=1ED8DA728BFDB3CF4C98EF696427A420D
- [16157-2] CEN TS 16157-2:2011 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 2: Location referencing, , https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:31062&cs=11C587676AF23E874354108C5B94B99F6
- [16157-3] CEN TS 16157-3:2011 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 3: Situation Publication, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:31063&cs=154597D857428FFFF988477C51088FE30
- [16157-4] CEN TS 16157-4:2014 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 4: Variable Message Sign (VMS) Publications, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:37597&cs=1046FF87225820240FCDAE9E4EF3118EB
- [16157-5] CEN TS 16157-5:2014 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 5: Measured and elaborated data publications, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:37966&cs=110BD77D7F2B9A32FB04403D101009B89
- [16157-6] CEN TS 16157-6:2015 Intelligent transport systems - DATEX II data exchange specifications for traffic management and information - Part 6: Parking Publications, https://standards.cencenelec.eu/dyn/www/f?p=CEN:110:0:::FSP_PROJECT:40220&cs=1084702C54C56E3547969771C8305250B
- [d2ref] DATEX II Current version/Reference set, <https://docs.datex2.eu/downloads/>.
- [d2supp] DATEX II Current version/Supporting, <https://docs.datex2.eu/>.